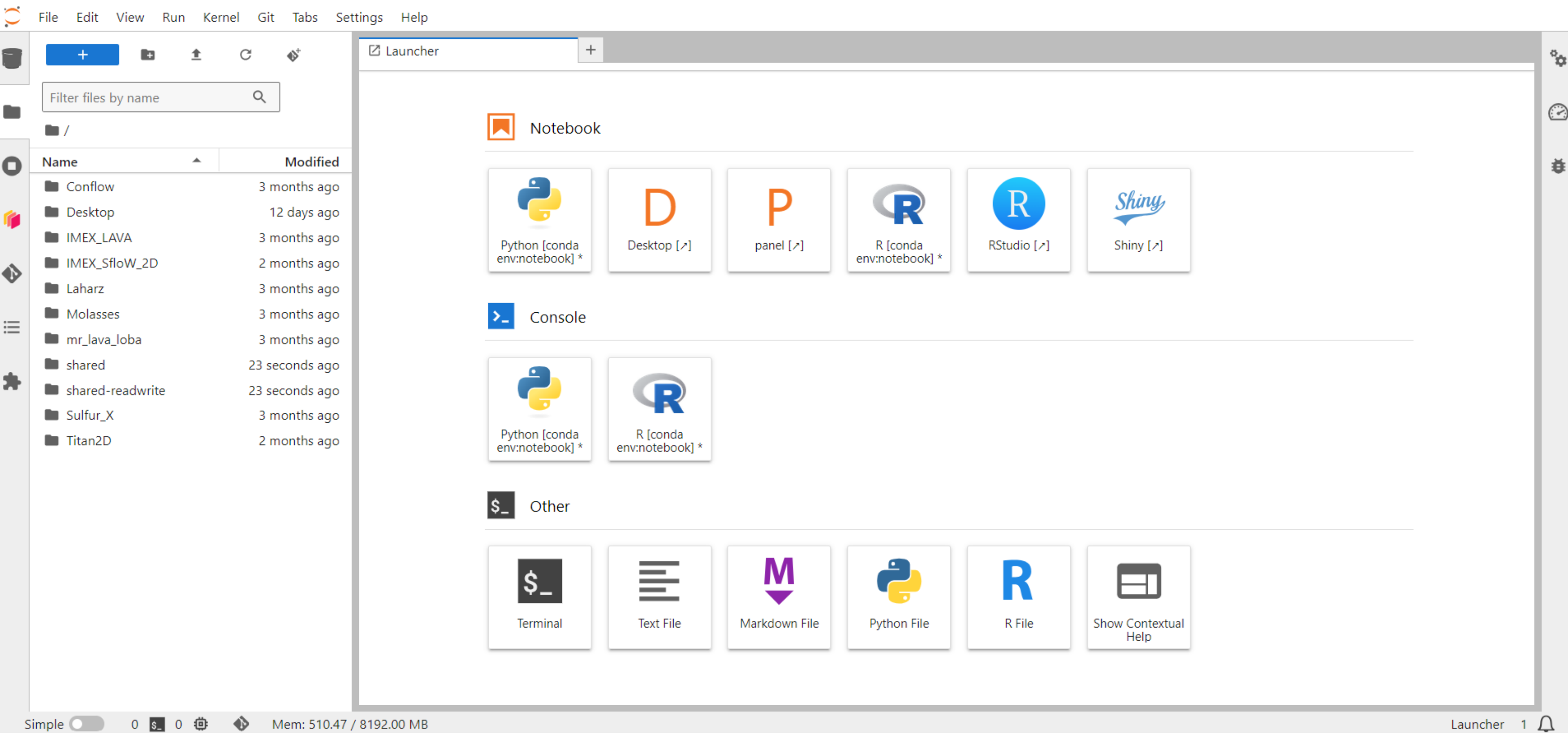
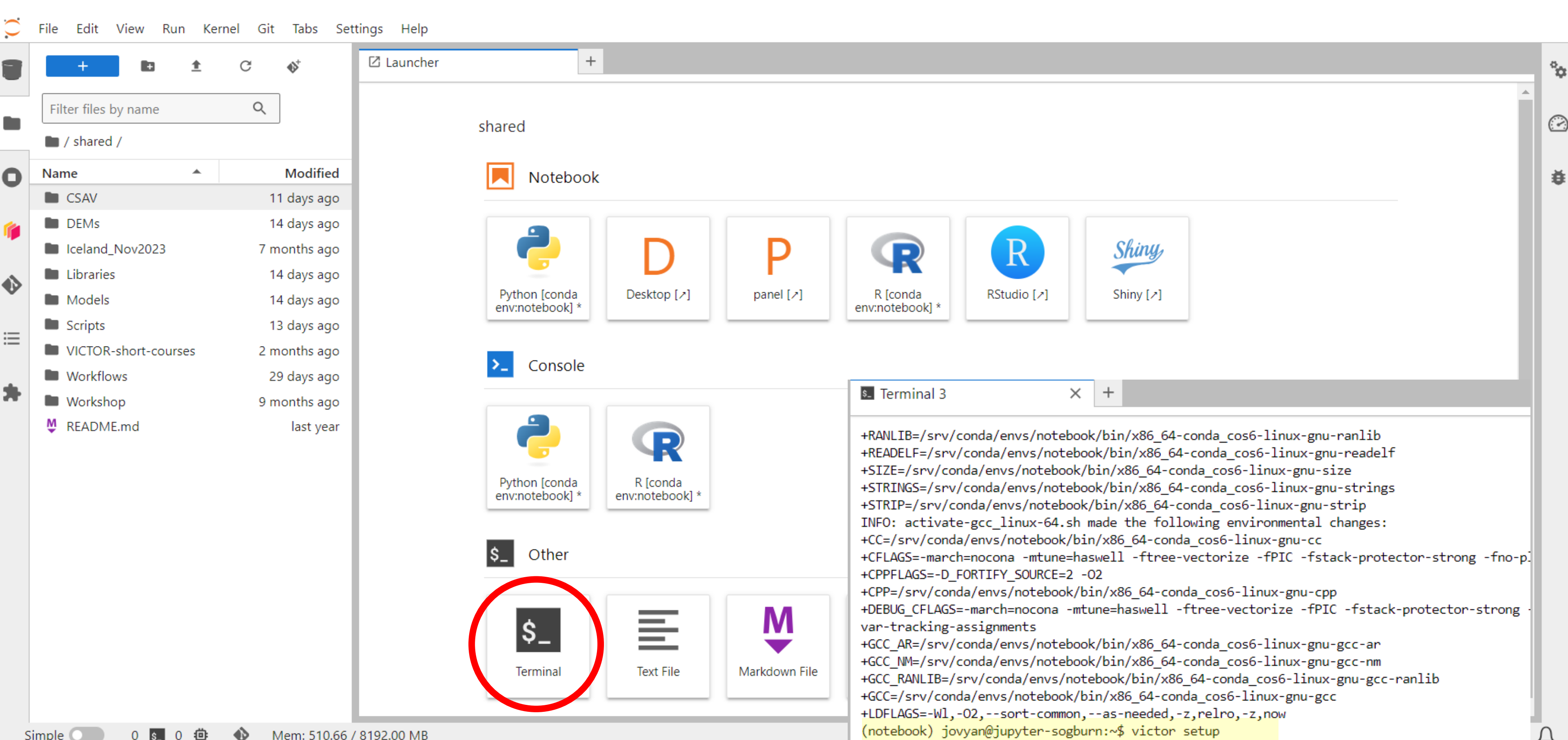
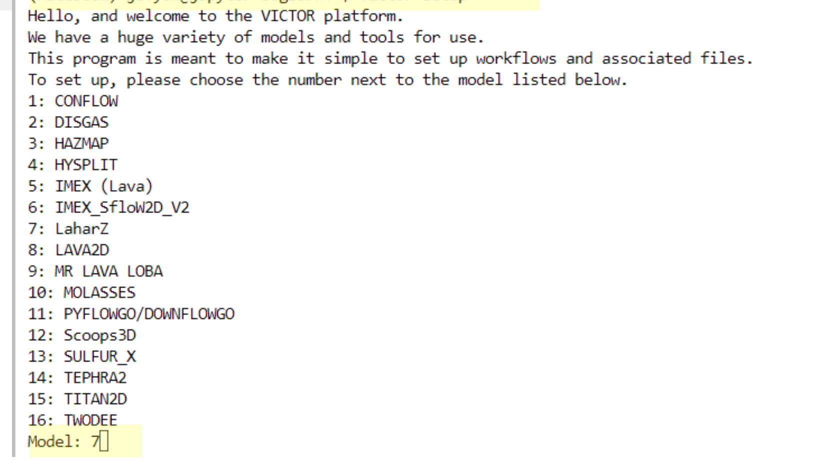
Intro to GIS
Exercise:
QGIS



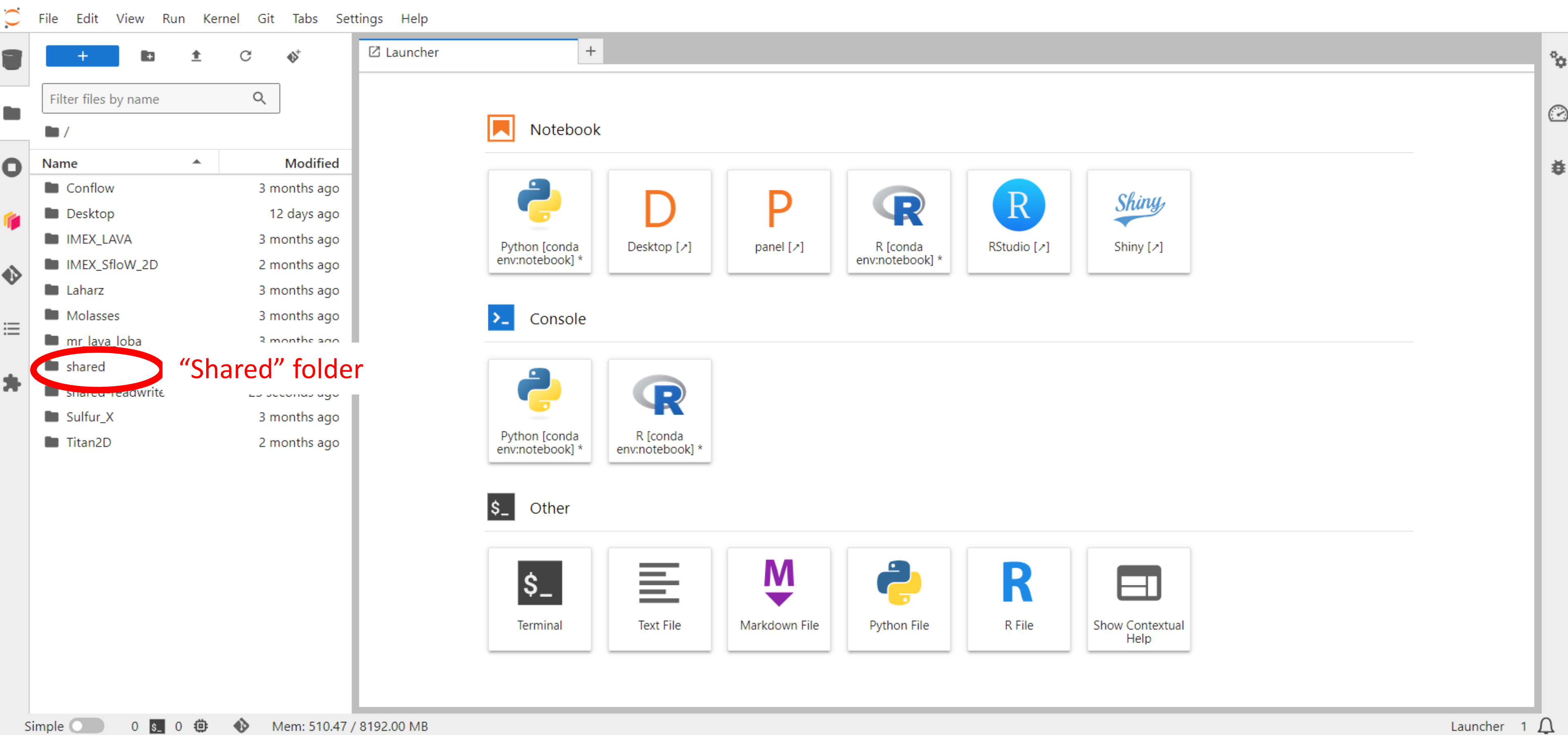
- When you login to VICTOR, your screen will look something like this
- Your files are on the left panel
- Your “Launcher” has useful icons

Tip: If you lose your “Launcher” screen, click the “plus” icon at the top left for “New Launcher”

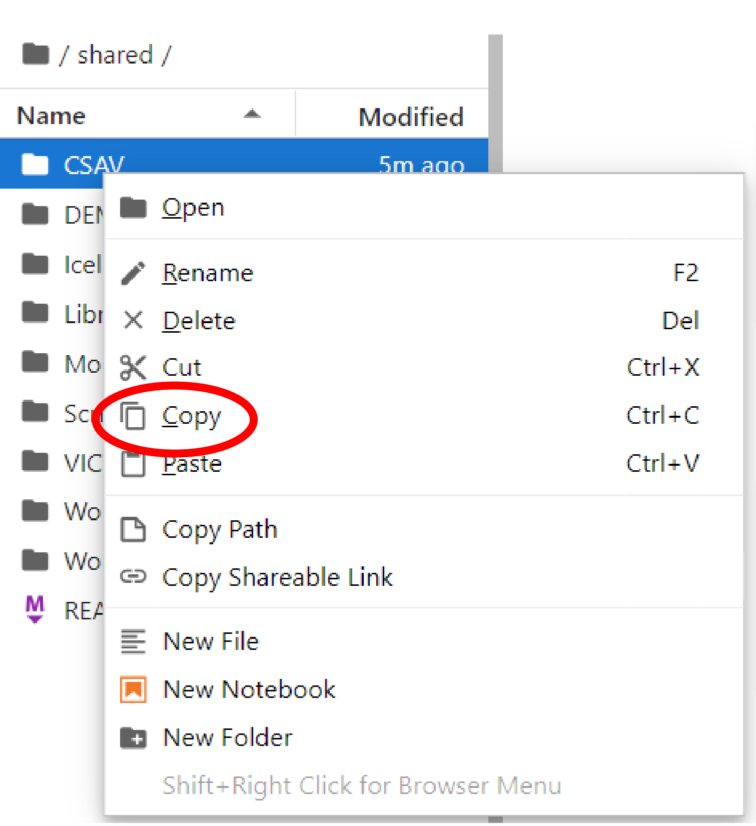


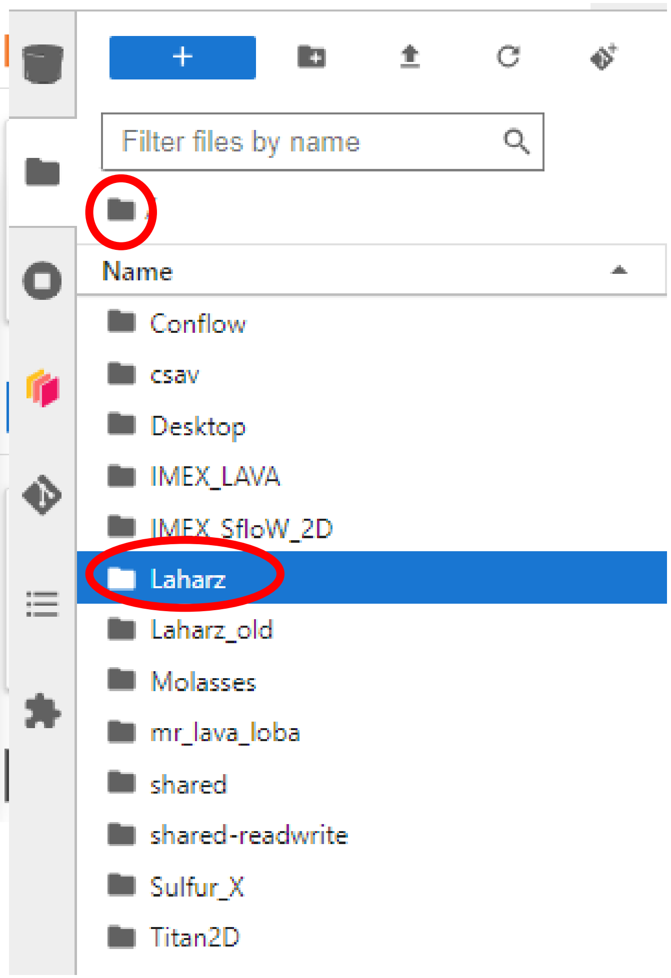


- We want to add the Laharz folder to your home workspace
- Click the terminal
- Type: victor setup
- Hit enter on your keyboard
- Type: 7
- Hit enter
- Close the terminal

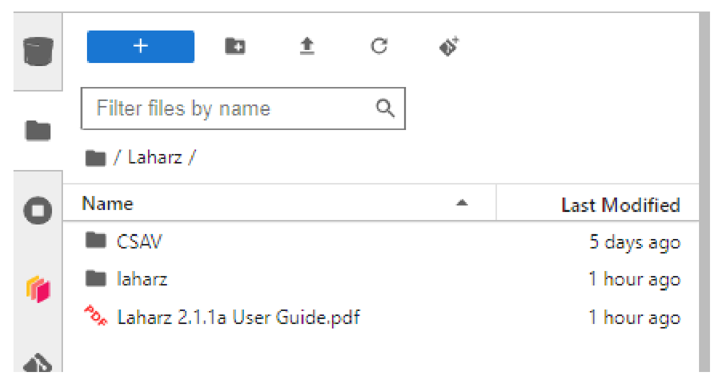


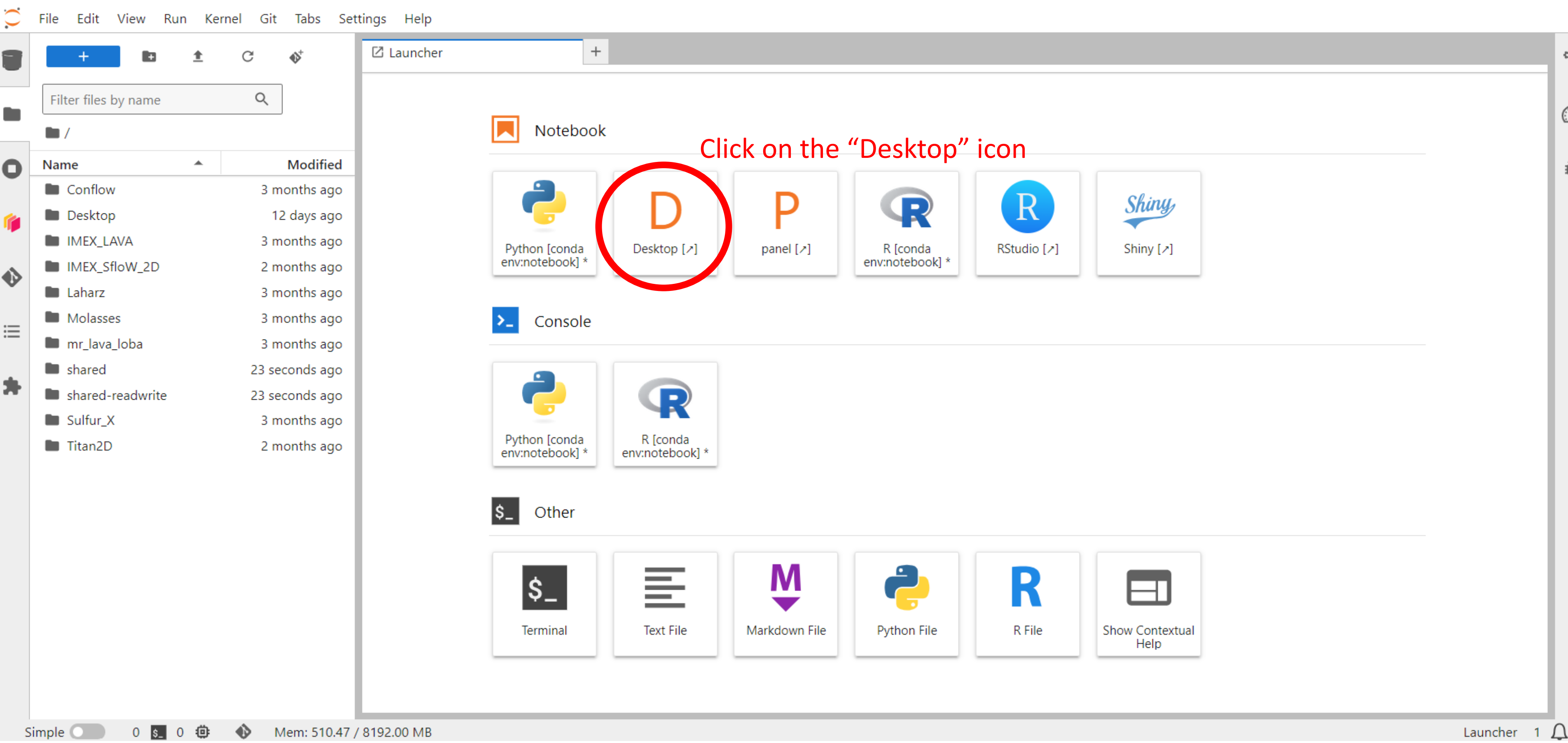
- Double click the “shared” folder

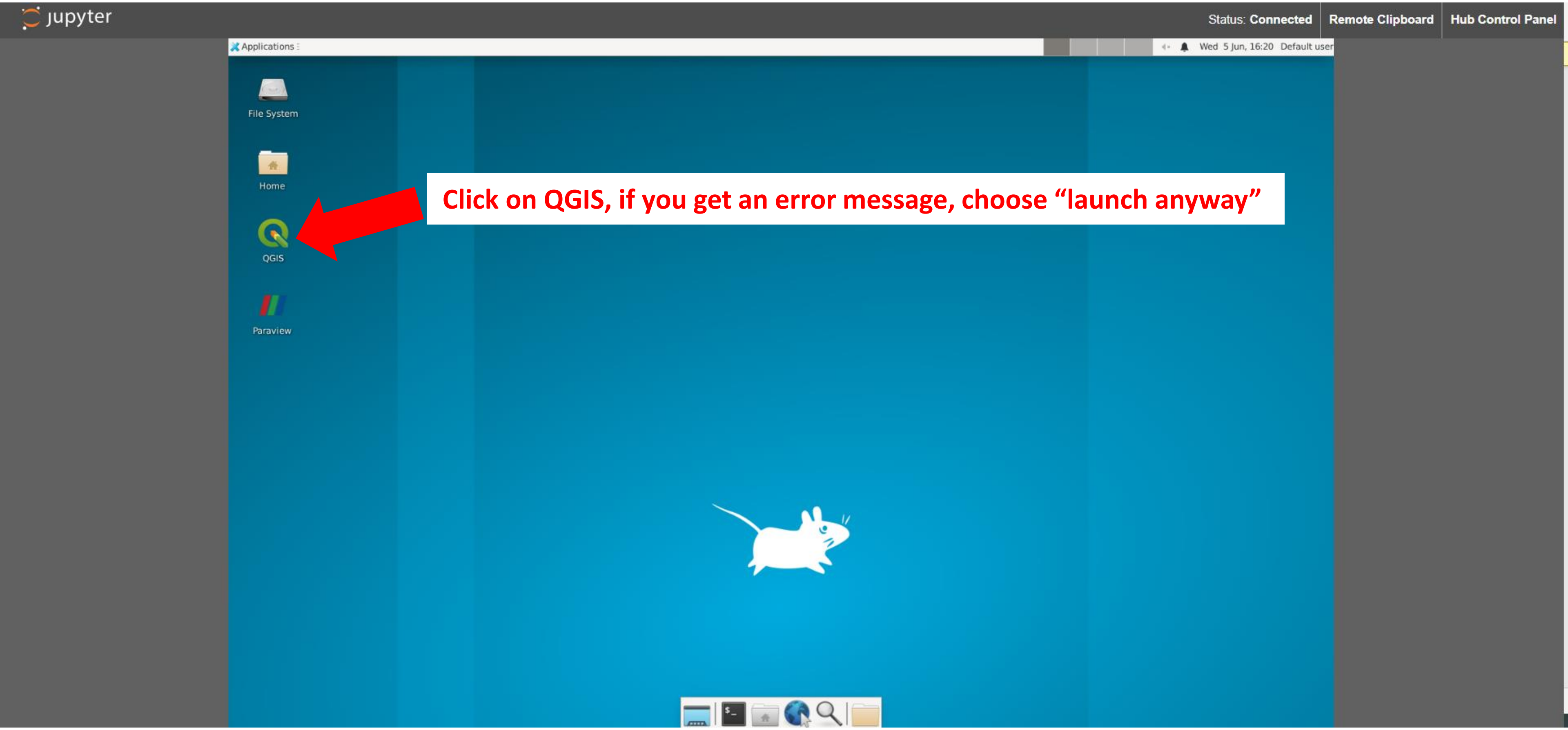


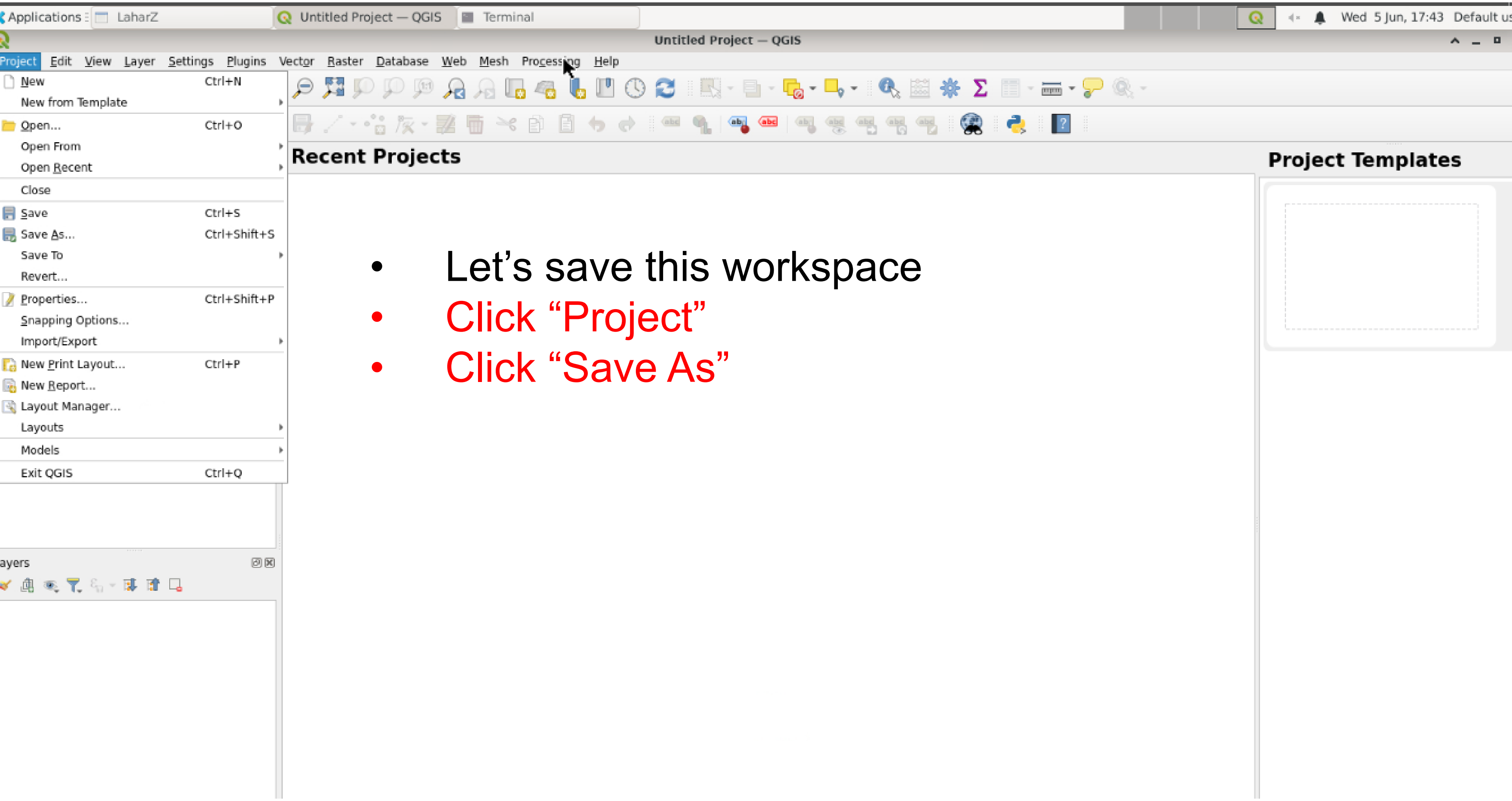


- Right click on the “CSAV” folder
- Click “Copy”
- Navigate to your home folder (click the folder icon)
- Navigate to the “Laharz” folder
- Paste the “CSAV” folder there

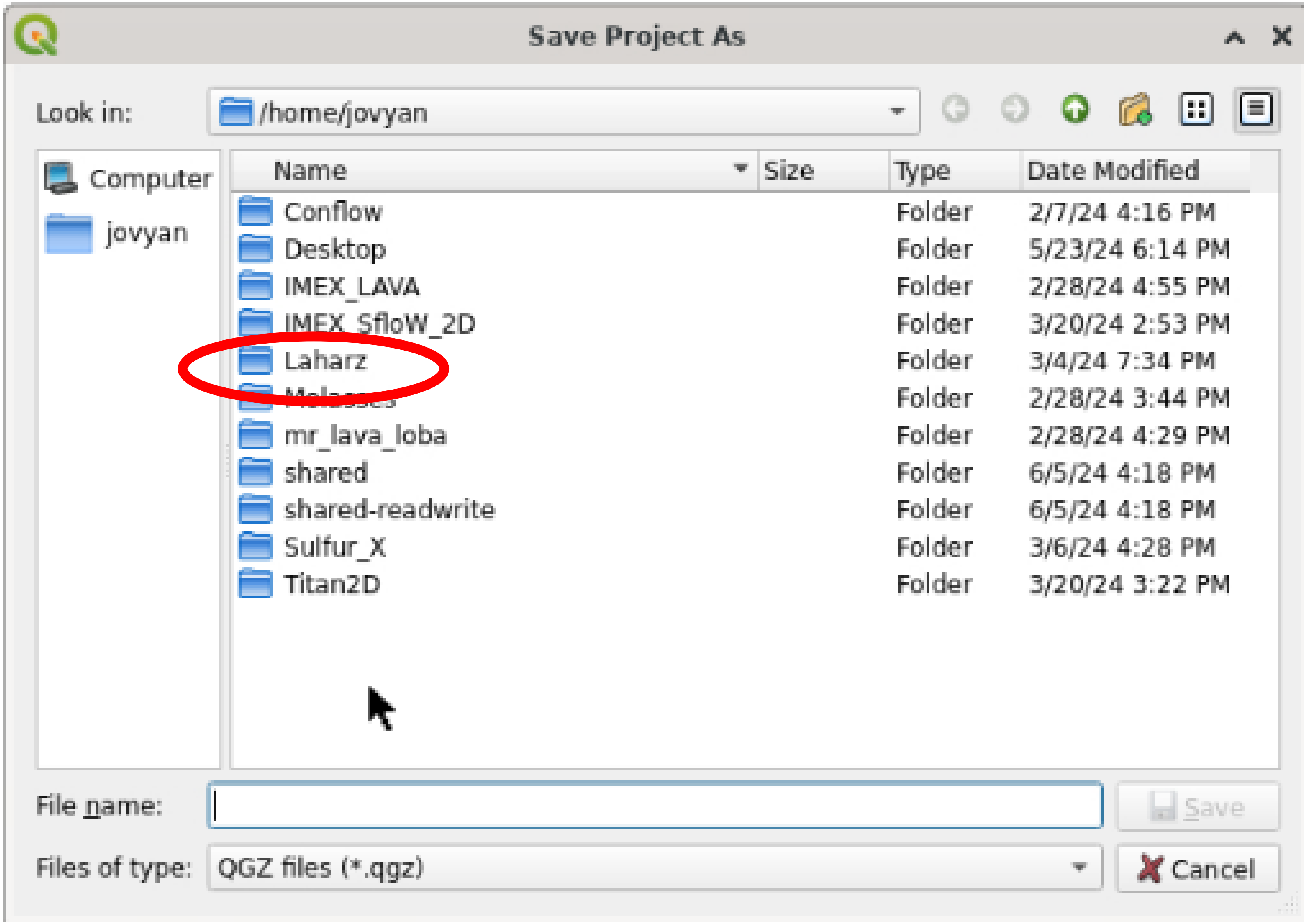



Click on the “Desktop” icon
File
Edit
View
Run
Kernel
Git
Tabs
Settings
Help
Launcher
Filter files by name
Name
Modified
Conflow 3 months ago
Desktop 12 days ago
IMEX_LAVA 3 months ago
IMEX_SfloW_2D 2 months ago
Laharz 3 months ago
Molasses 3 months ago
mr_lava_loba 3 months ago
shared 23 seconds ago
shared-readwrite 23 seconds ago
Sulfur_X 3 months ago
Titan2D 2 months ago
Notebook
Python [conda env:notebook] *
Desktop [↗]
panel [↗]
R [conda env:notebook] *
RStudio [↗]
Shiny [↗]
Console
Python [conda env:notebook] *
R [conda env:notebook] *
Other
Terminal
Text File
Markdown File
Python File
R File
Show Contextual Help
Simple
Mem: 510.47 / 8192.00 MB
Launcher

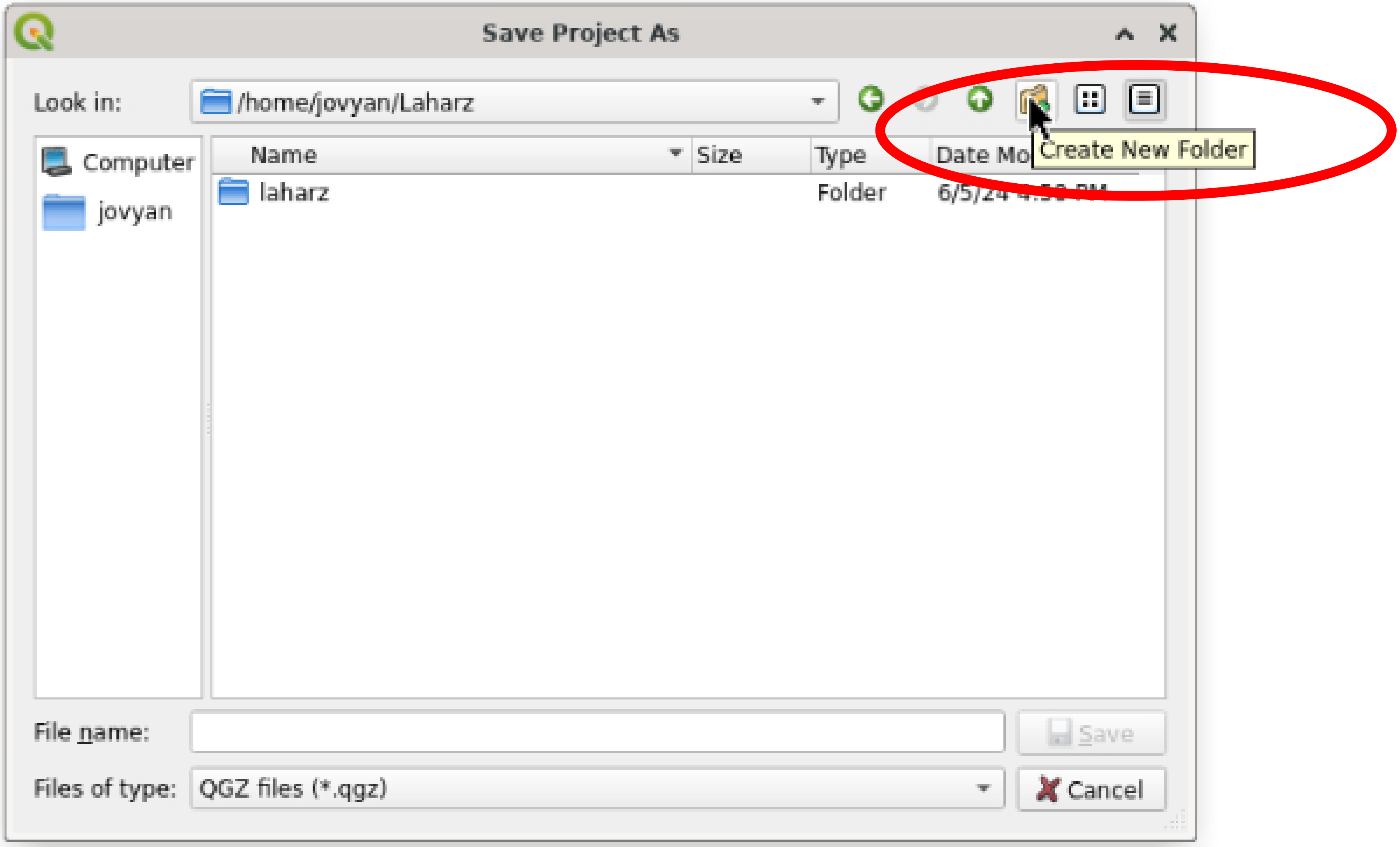

jupyter
Status: Connected
Remote Clipboard
Hub Control Panel
Applications
Wed 5 Jun, 16:20 Default user
File System
Home
QGIS
Paraview
Click on QGIS, if you get an error message, choose “launch anyway”



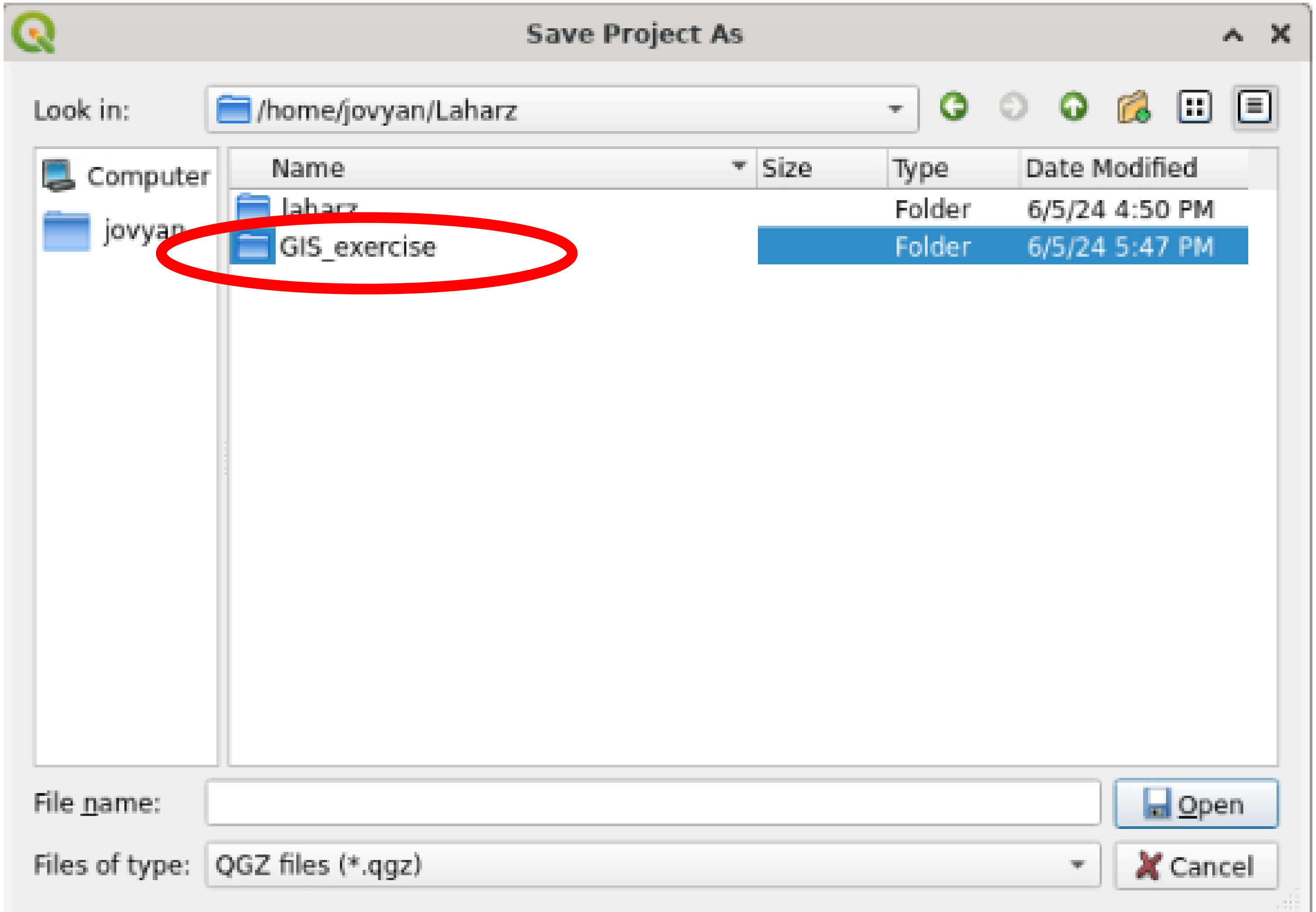
- Let’s save this workspace
- Click “Project”
- Click “Save As”



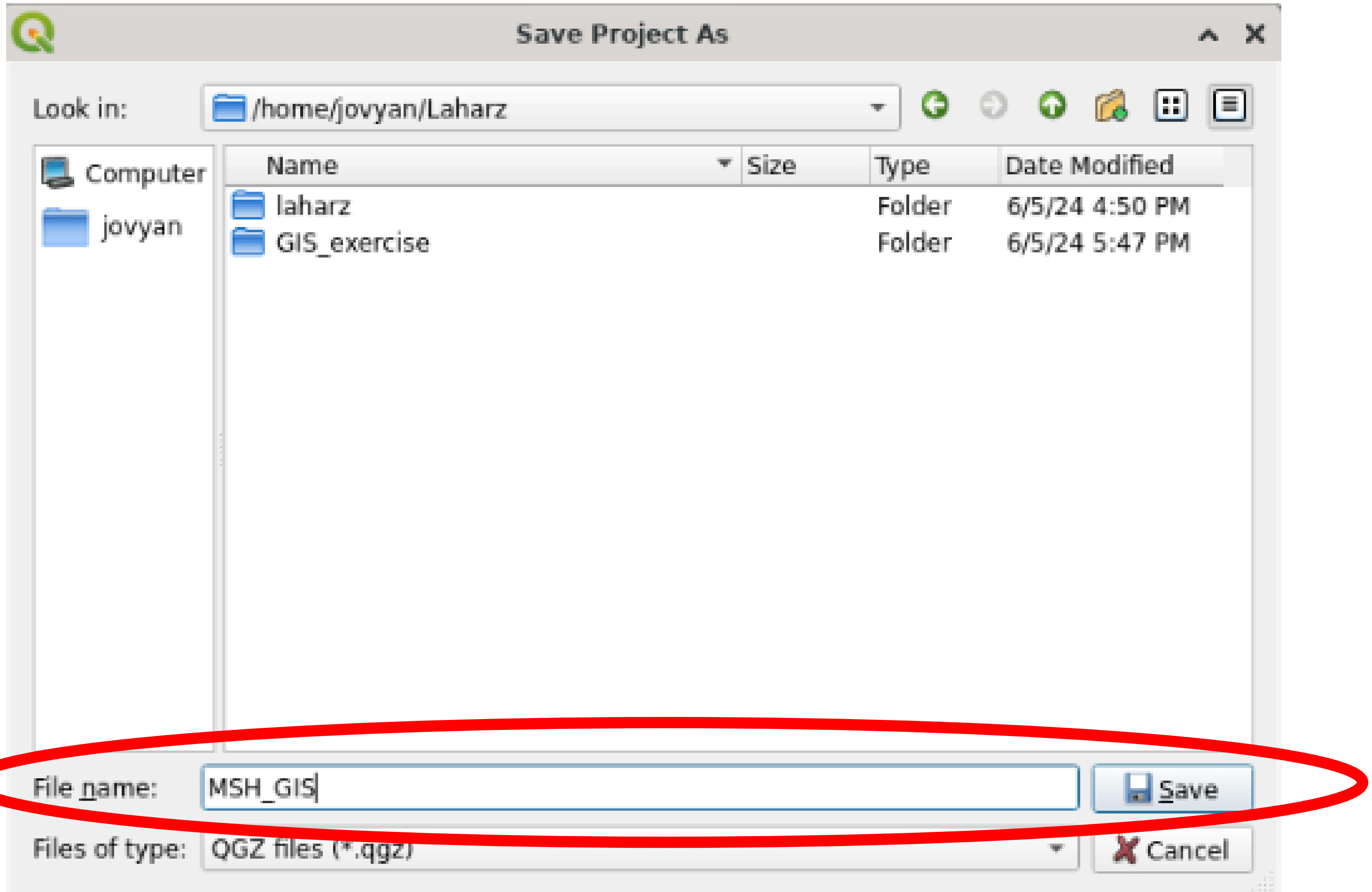
- Navigate to the “Laharz” folder



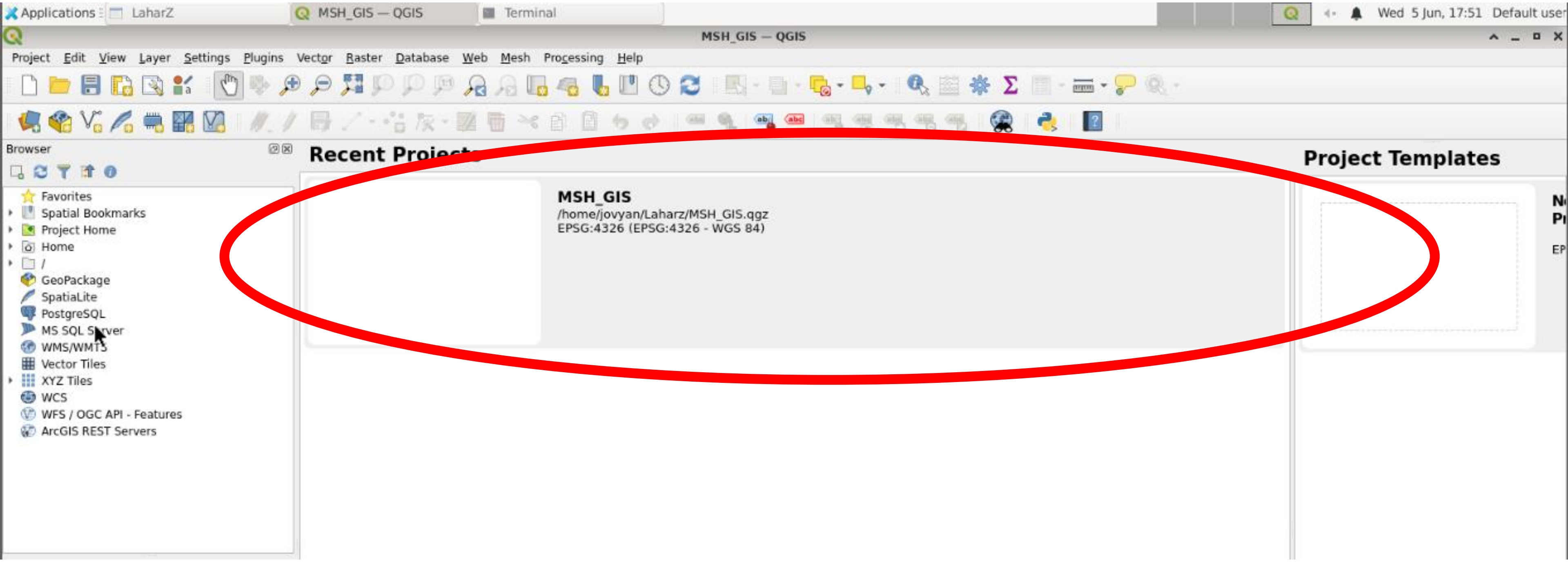
- Click “Create New Folder”



- Name the new folder "GIS_exercise" or something similar



- In the “File name” box, type the name of your project
- I named this “MSH_GIS”
- Click “Save”



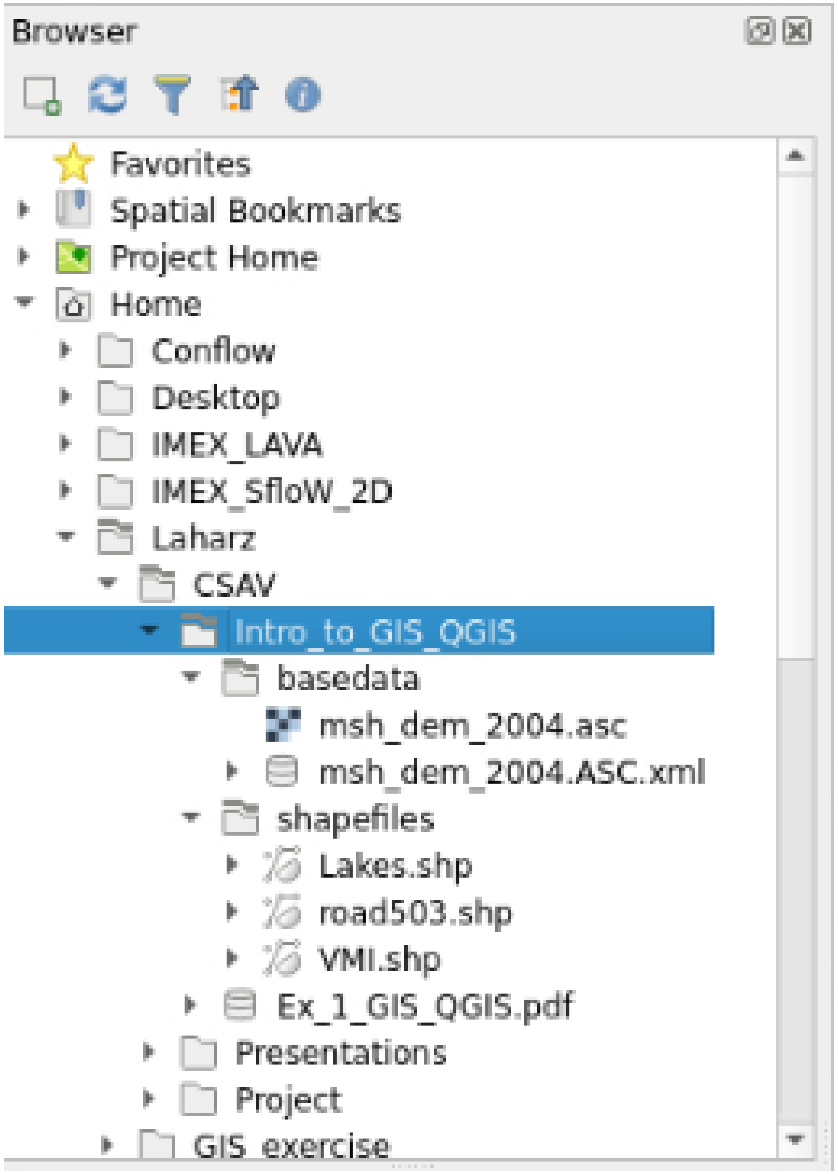
- Click on the new project

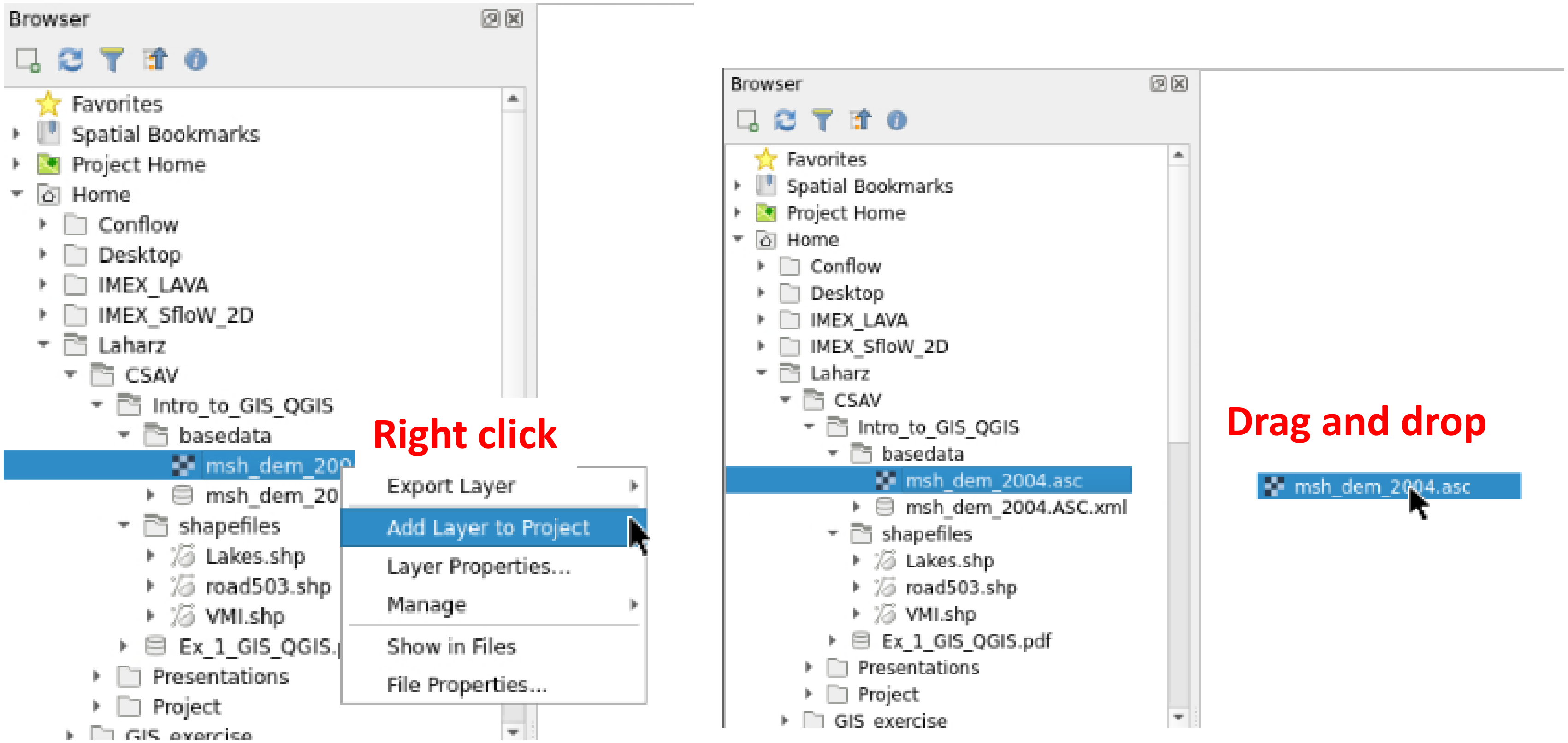
Exercise Goals

- **Add Instrument data set and background information for context**
- View Instrument attribute table
- Select instruments by drawing rectangle
- Select instruments by their attributes
- Assign colors to instrument type
- Add a legend to describe the meaning of colors

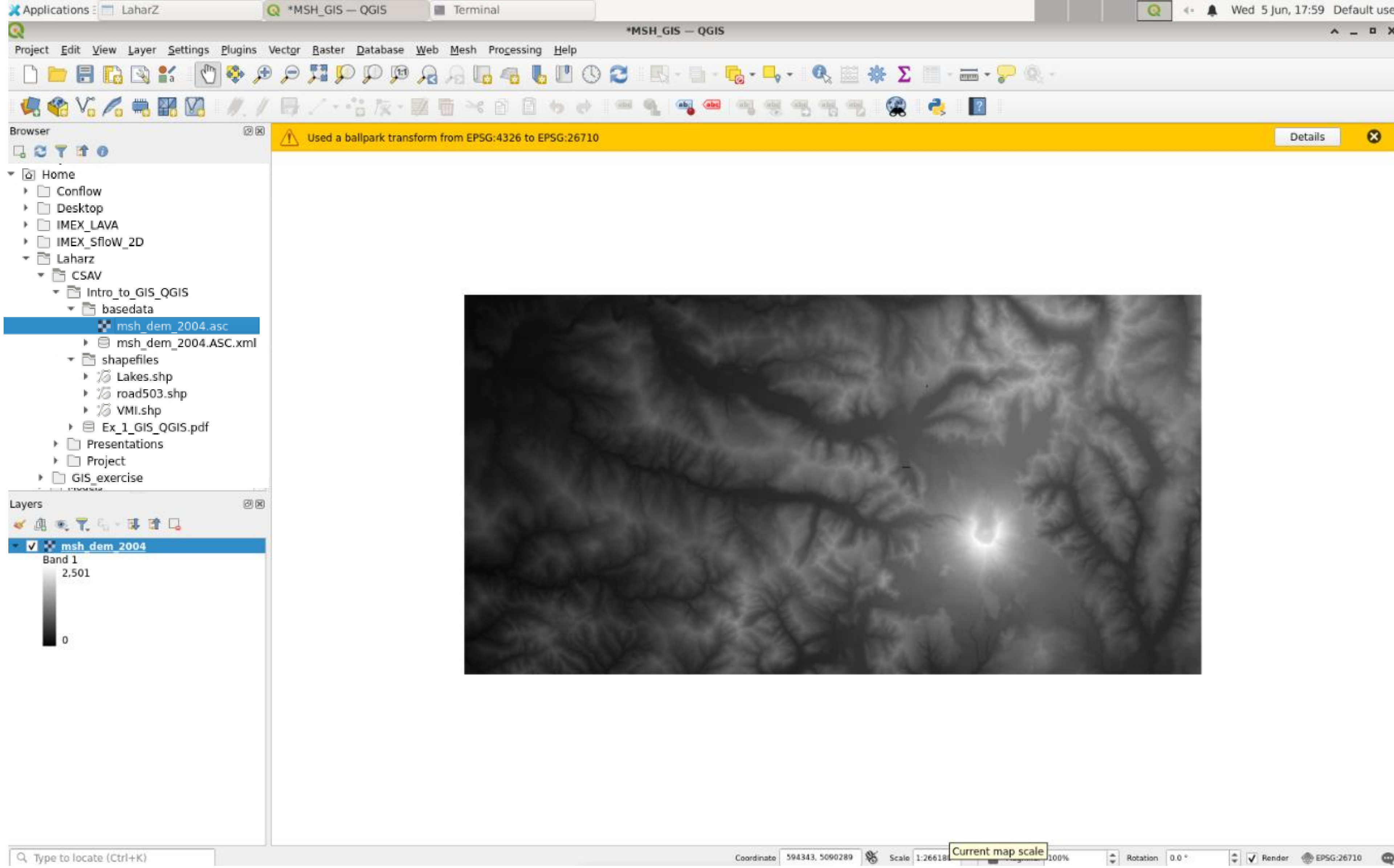
Navigate to files

- In the “Browser” panel of QGIS, navigate to the Laharz/CSAV/Ex_1_Intro_to_GIS_QGIS folder
- In this folder are two folders:
  - Basedata with the Mt. St. Helens 2004 DEM
  - Shapefiles with lake, road, and the volcano monitoring instrument (VMI) layers

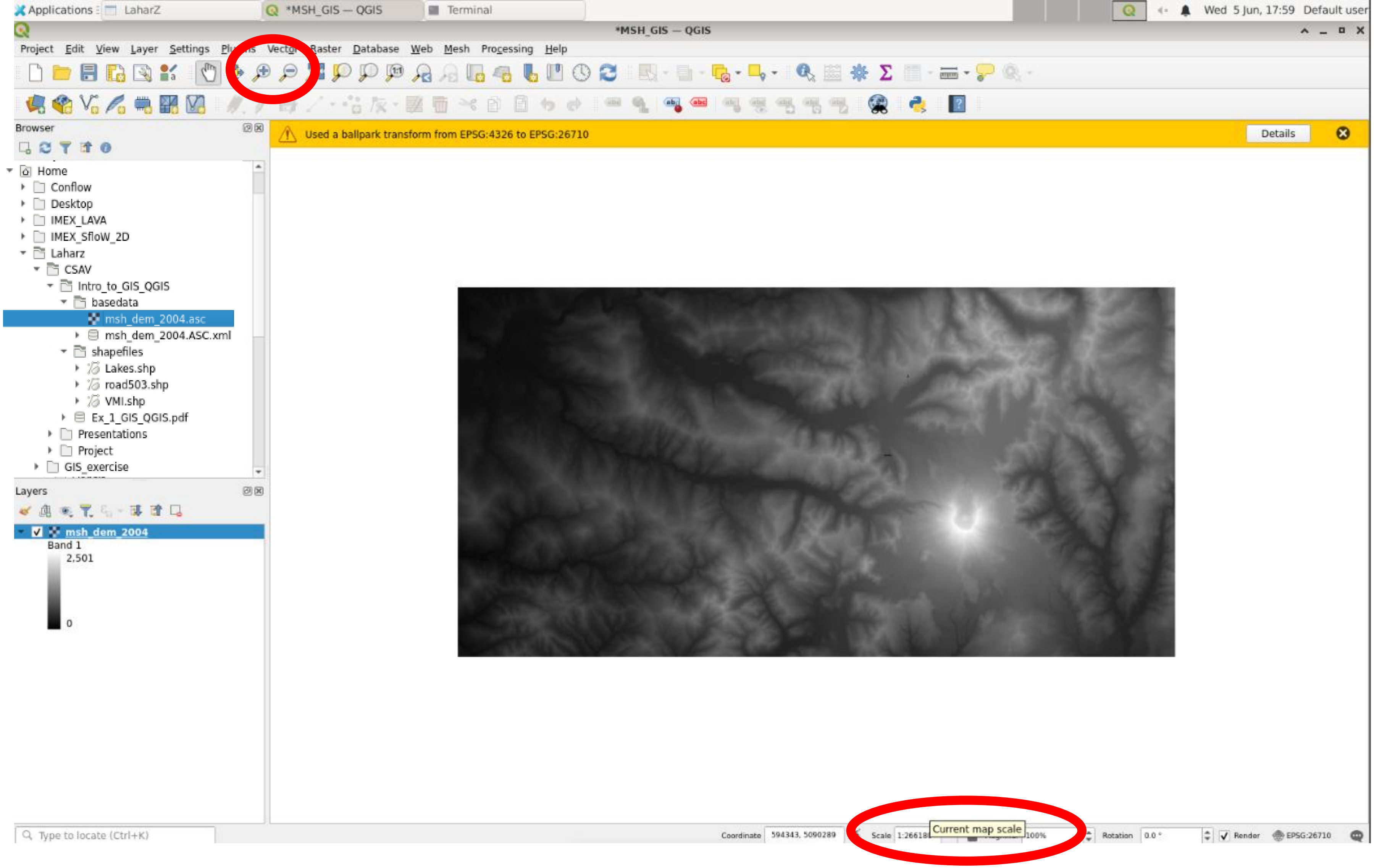




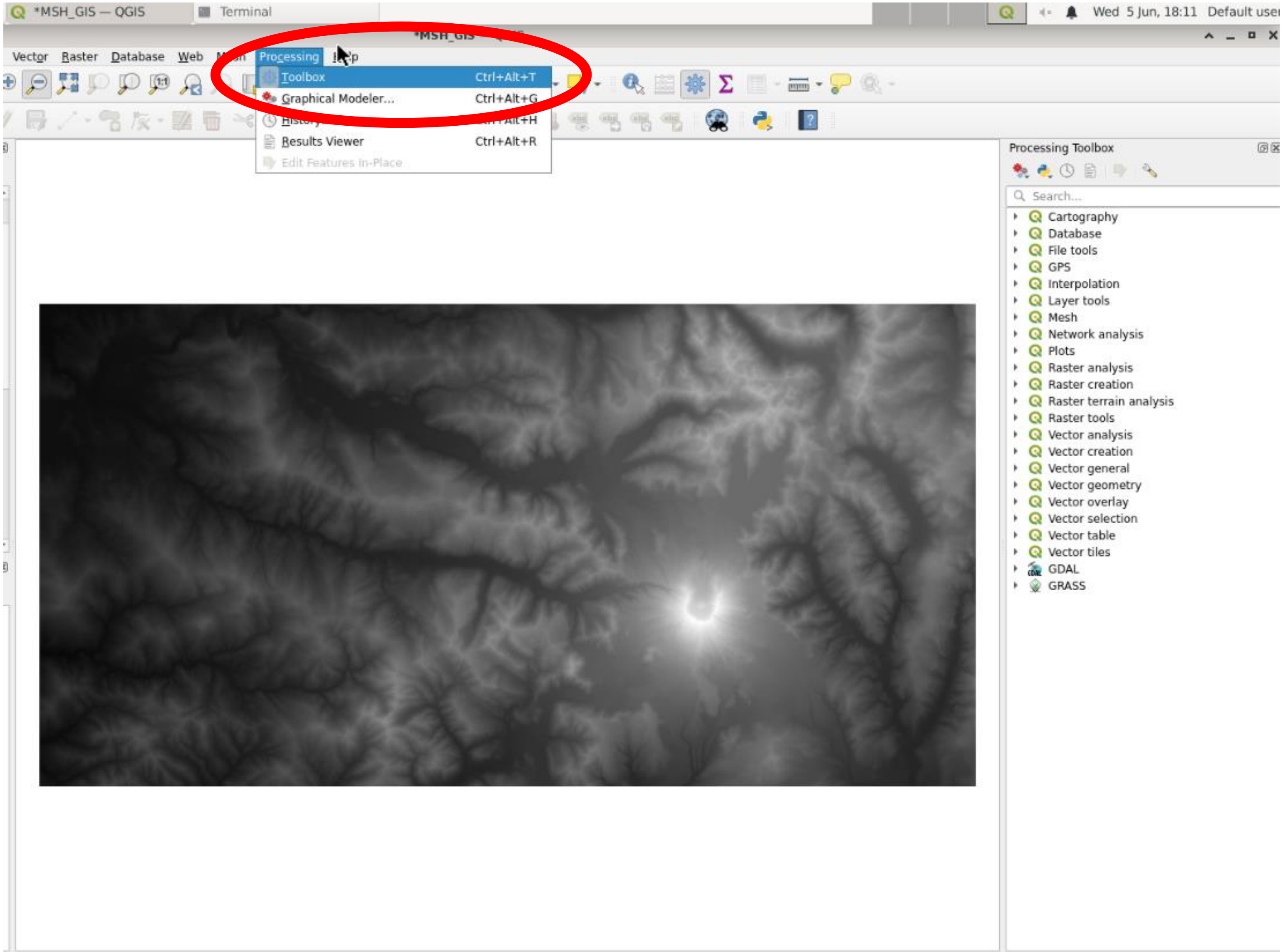
- Two ways to add layers:
  - Right click on the “msh_dem_2004.asc” file and choose “Add Layer to Project”
  - Drag the “msh_dem_2004.asc” layer and drop onto the workspace



- You should see the MSH DEM on the screen

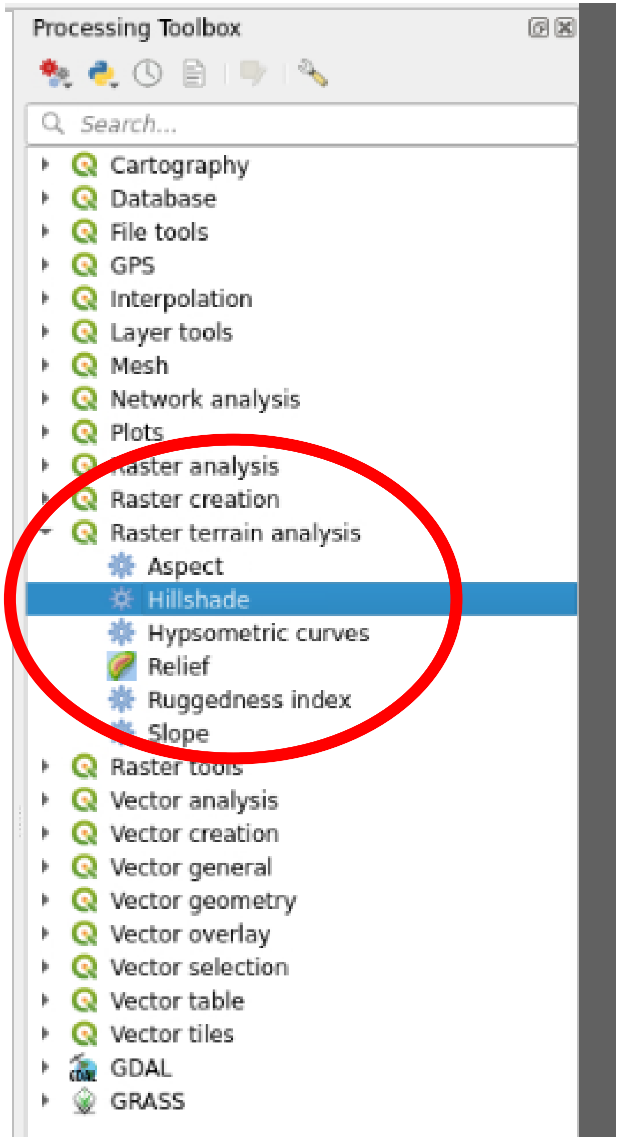


- You can zoom in and out 3 ways:
  - Use the mouse wheel
  - Click the magnifying glass icon, then click on the map
  - Edit the map scale manually at the bottom of the screen. Try 1:90000



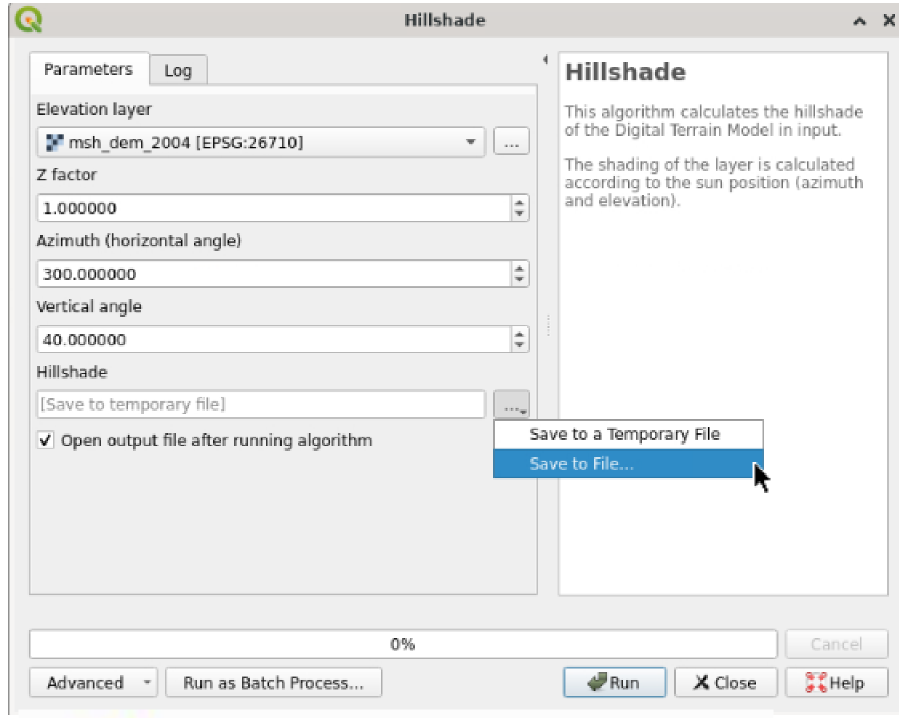
- Open the Processing Toolbox

- Let’s make a hillshade
  - A hillshade (or shaded-relief model) is a rendition of shadows across a DEM that helps our brains interpret what we see in DEMs.
  - Click the “Raster terrain analysis” menu
  - Double click the “Hillshade” tool
  - You can also use the “Search” field to search for tools!



Let’s make a hillshade

- This tool let’s you model a “sun” and the “shadows” it would produce on your DEM
- Use the default parameters
  - Z factor is vertical exaggeration (if your z units differ from your x and y units)
  - The azimuth is where the sun is relative to your map. The human brain expects the shadows to be modeled with the sun coming from the upper left (300°)
  - The vertical angle is how high in the sky the modeled sun is. We use the default 40°



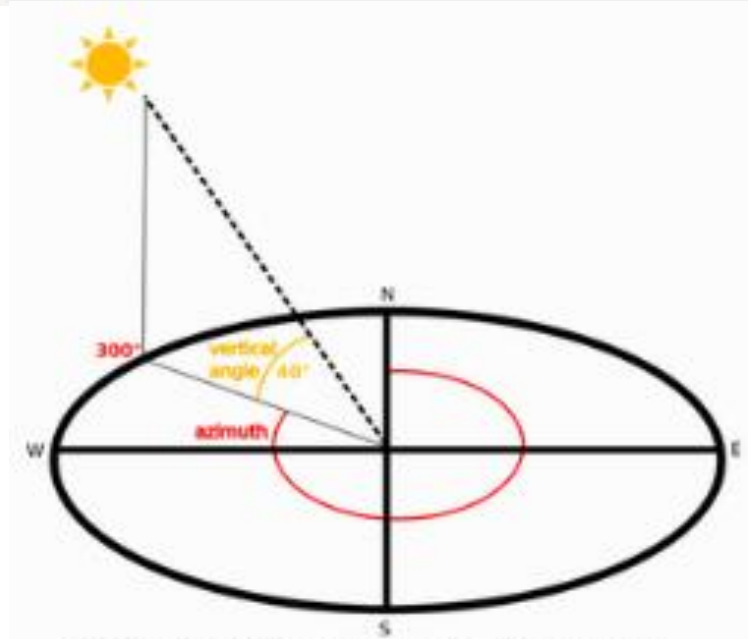
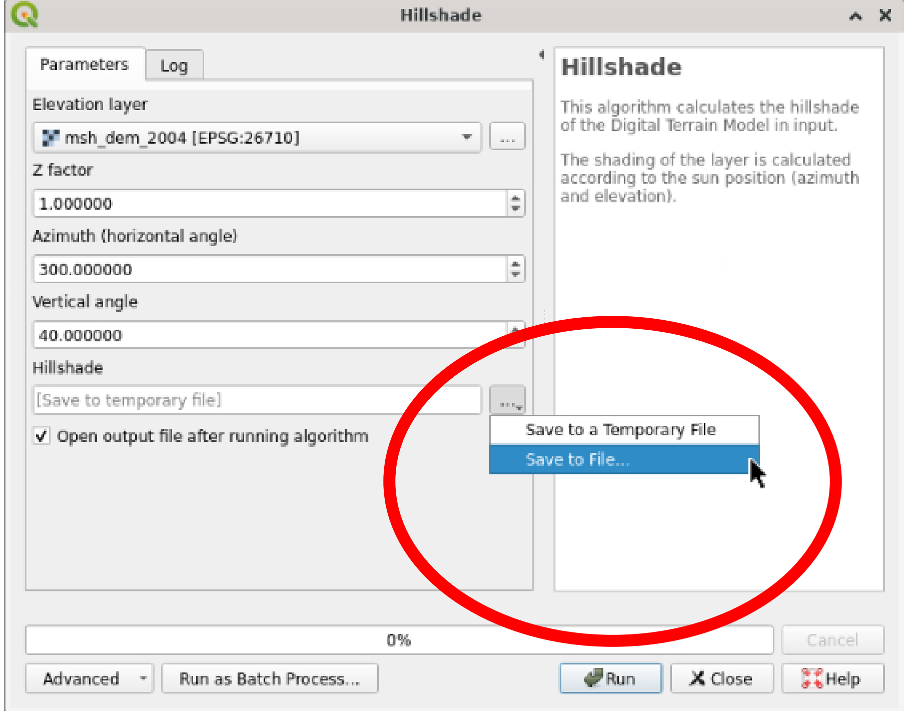
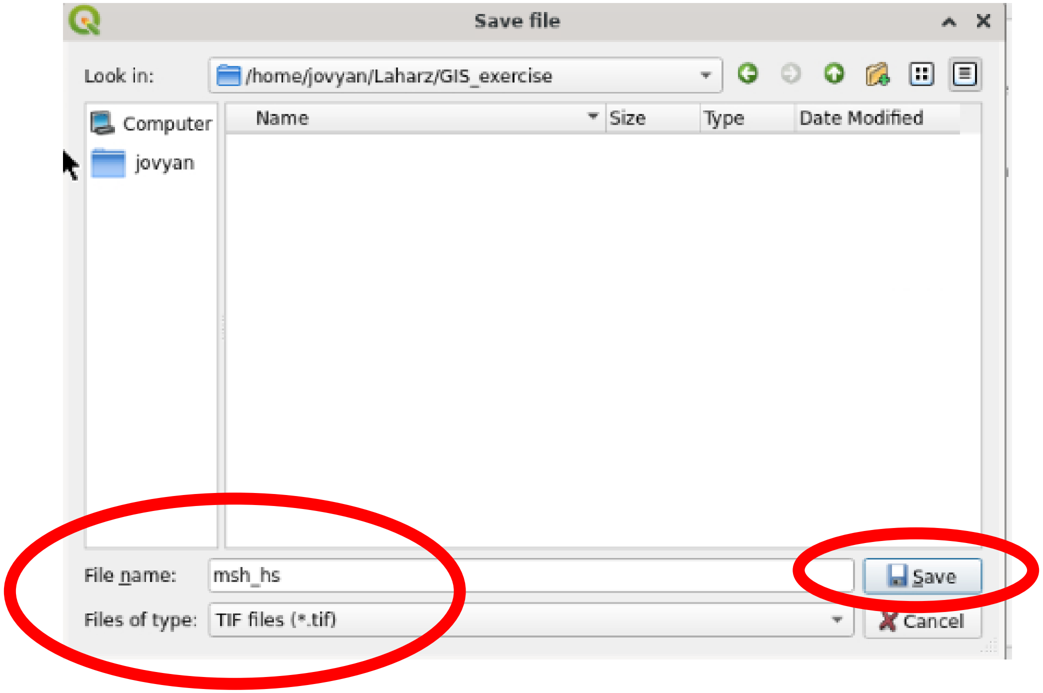


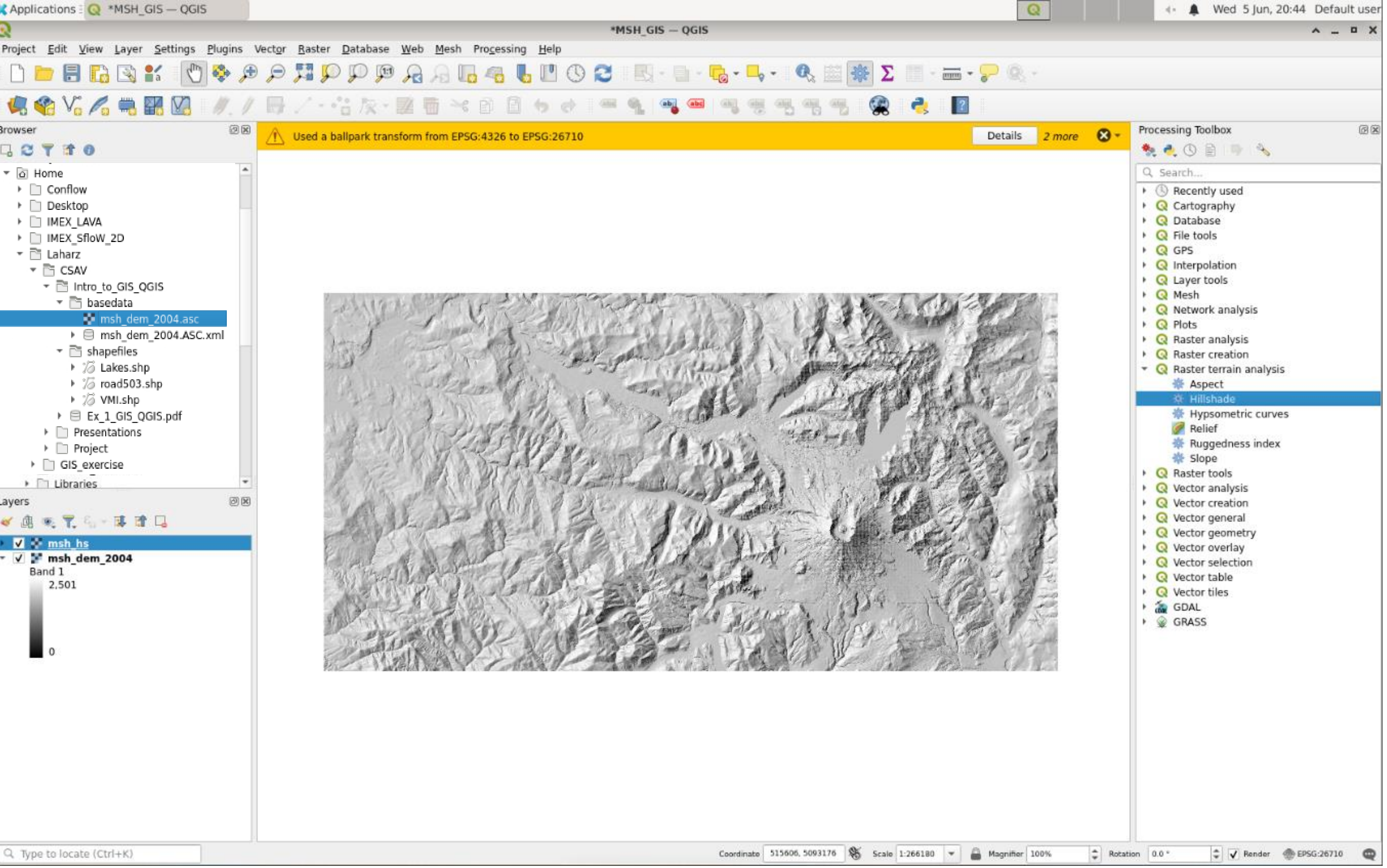
Fig. 28.32 *Azimuth and vertical angle*

Let’s make a hillshade

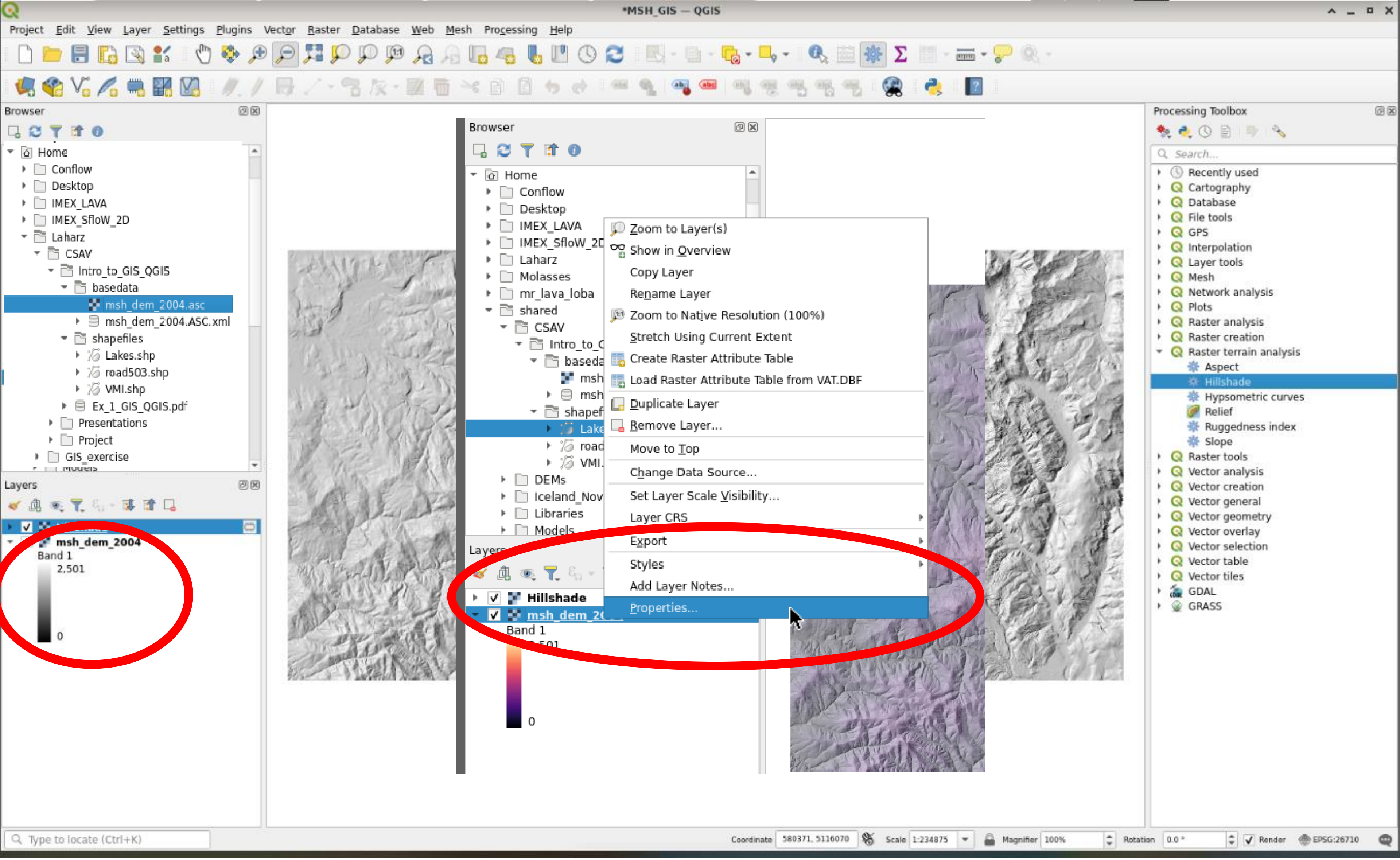
- Click the 3 dot menu next to the Hillshade filename box and select “Save to File”
- Navigate to Laharz/GIS_exercise
- Enter a filename. I used “msh_hs”
- Click “Save”
- Click “Run”



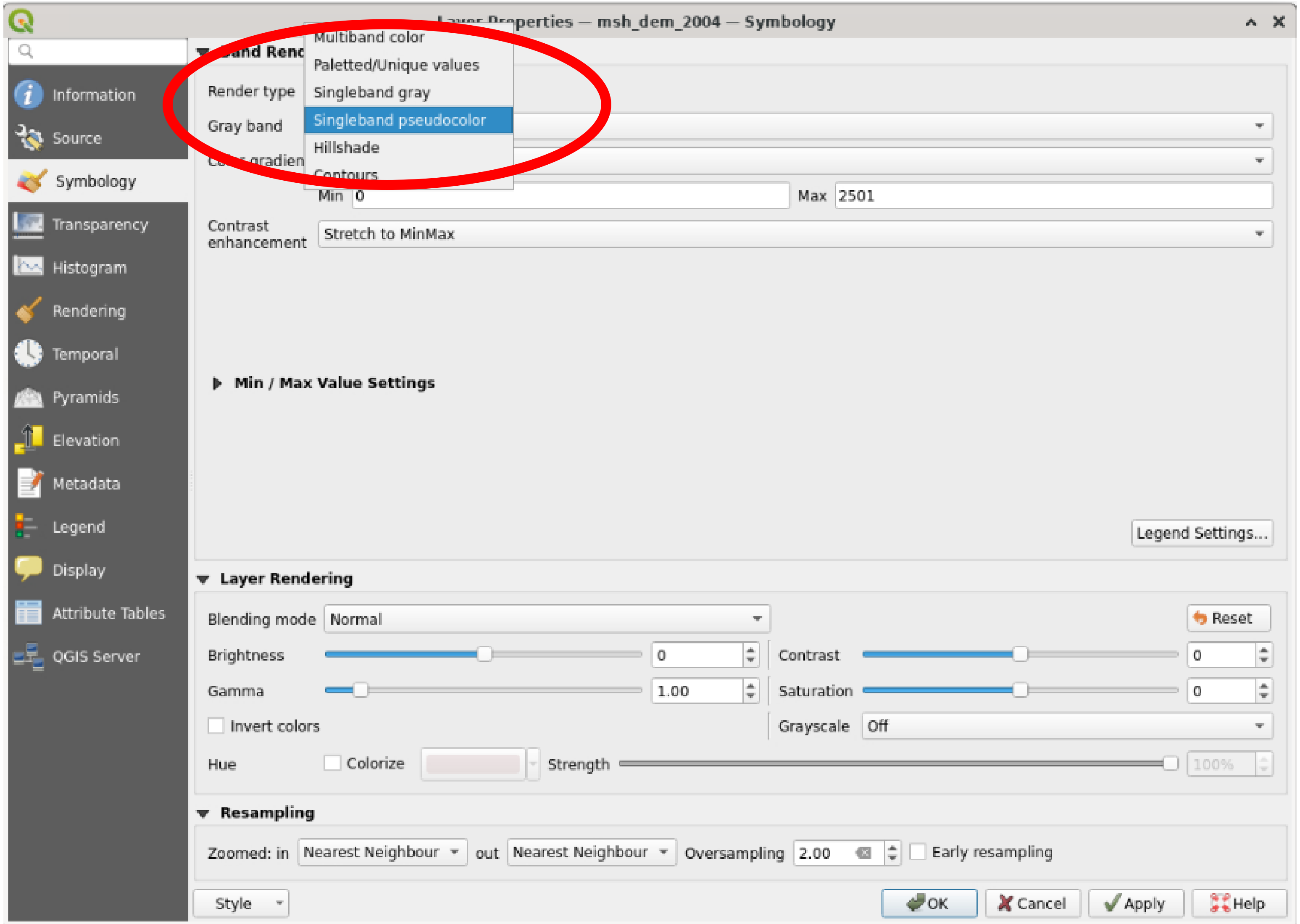




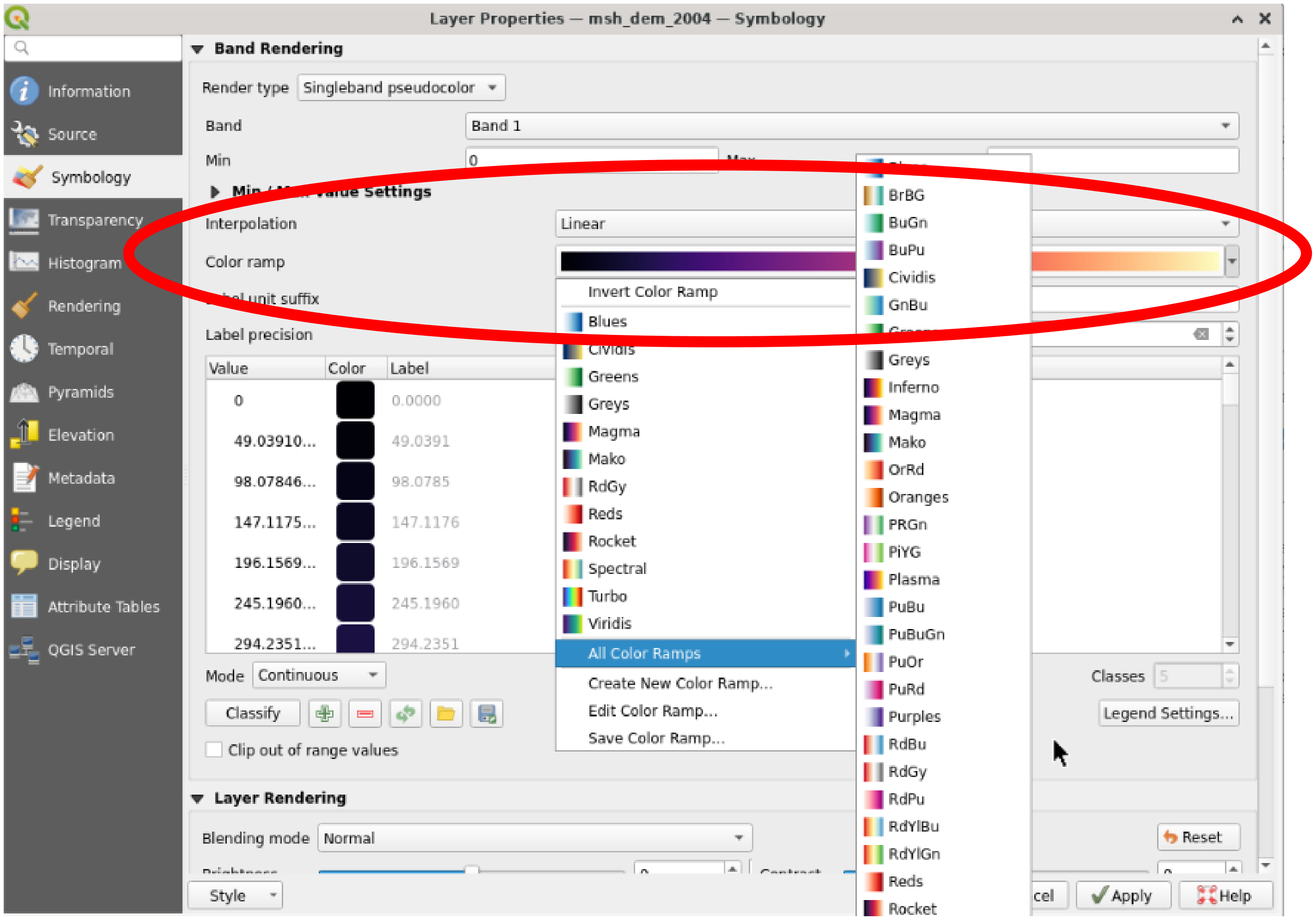
- We can change the symbology of the DEM and the transparency of the hillshade if we want
- Hillshades (shaded relief) use shades of grey 0-254. DEMs use a stretched color spectrum from lowest to highest elevation



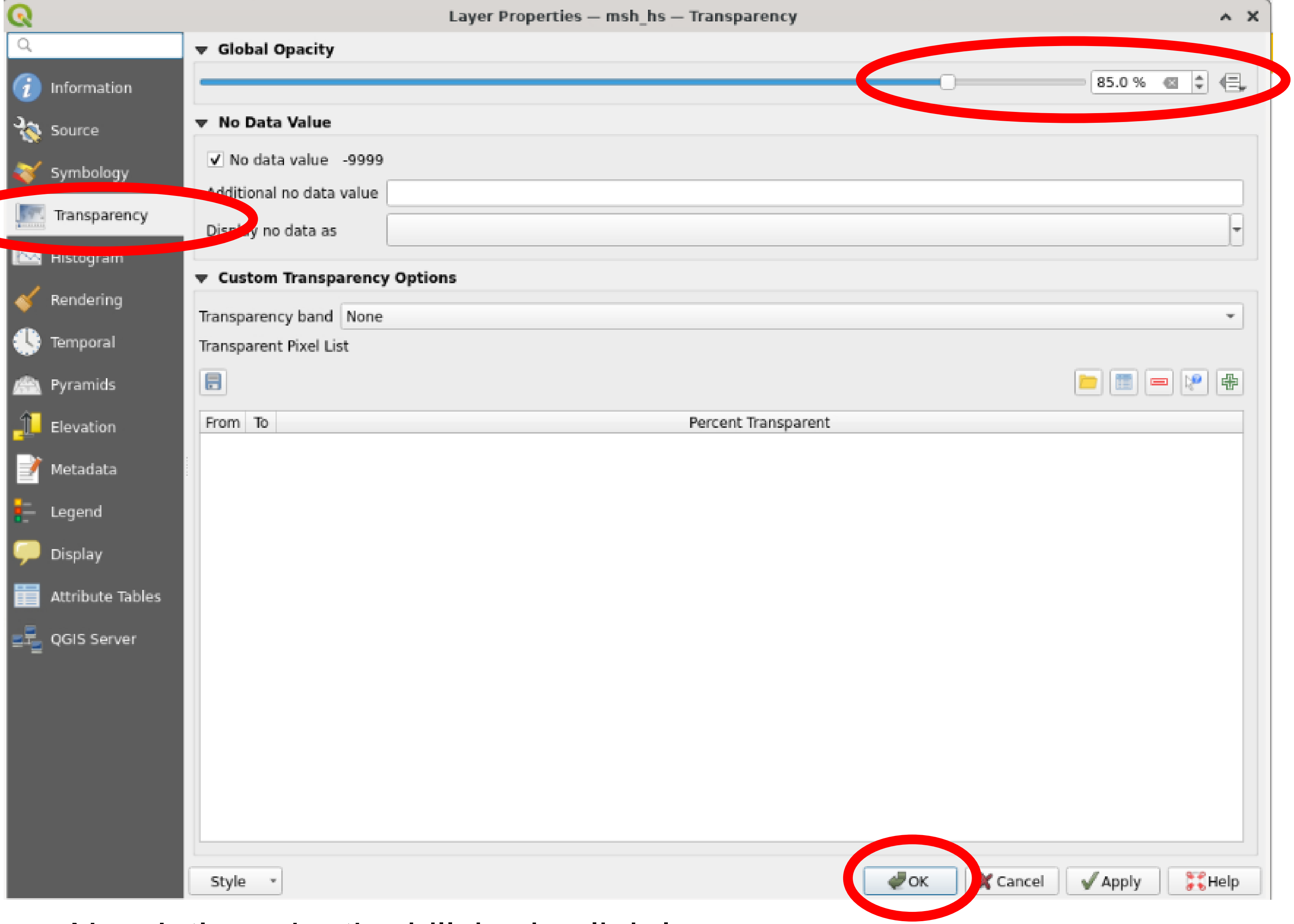
- Double click on the msh_dem_2004 layer to open the layer properties
- OR right click on the “msh_dem_2004” layer and choose “Properties”



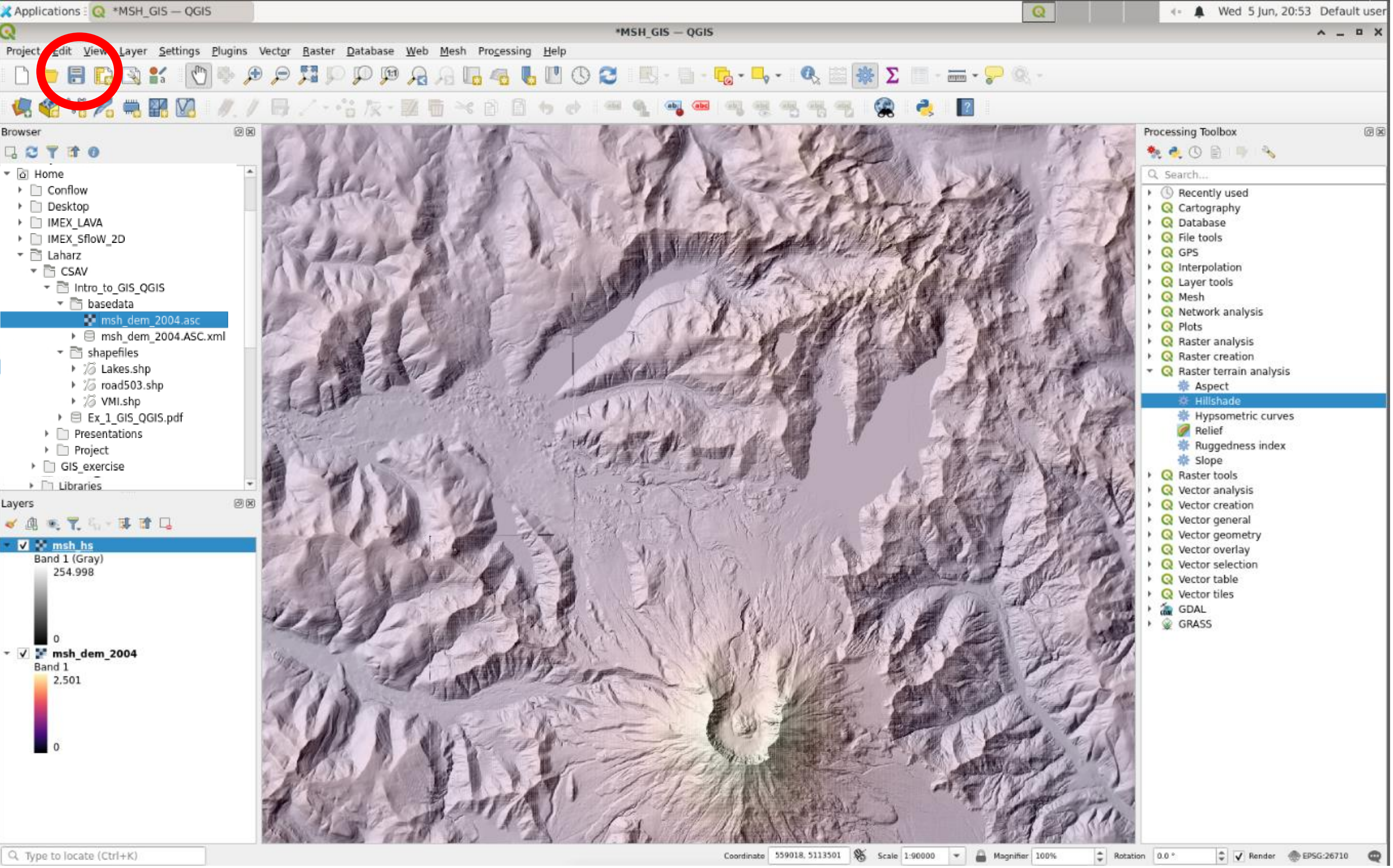
- Change the “Render type” to “Singleband pseudocolor”



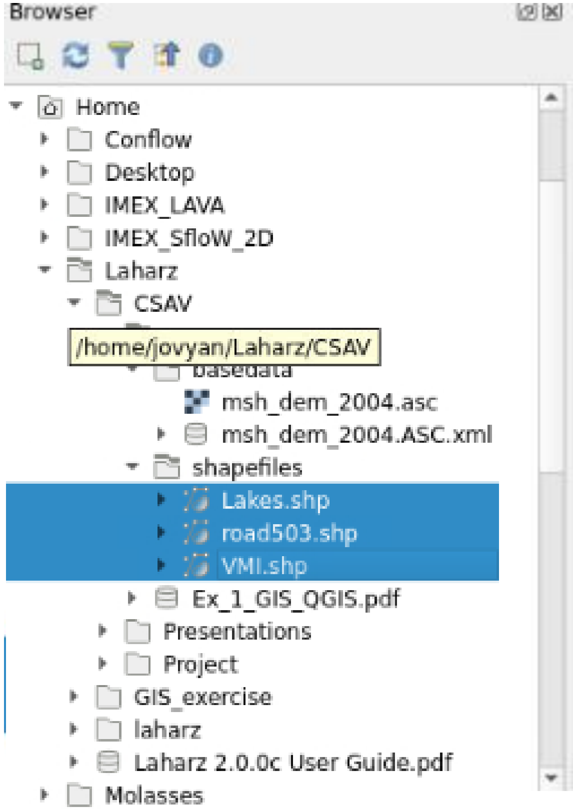
- Choose a color ramp
- I chose the “Magma” color ramp
- Click “Ok”



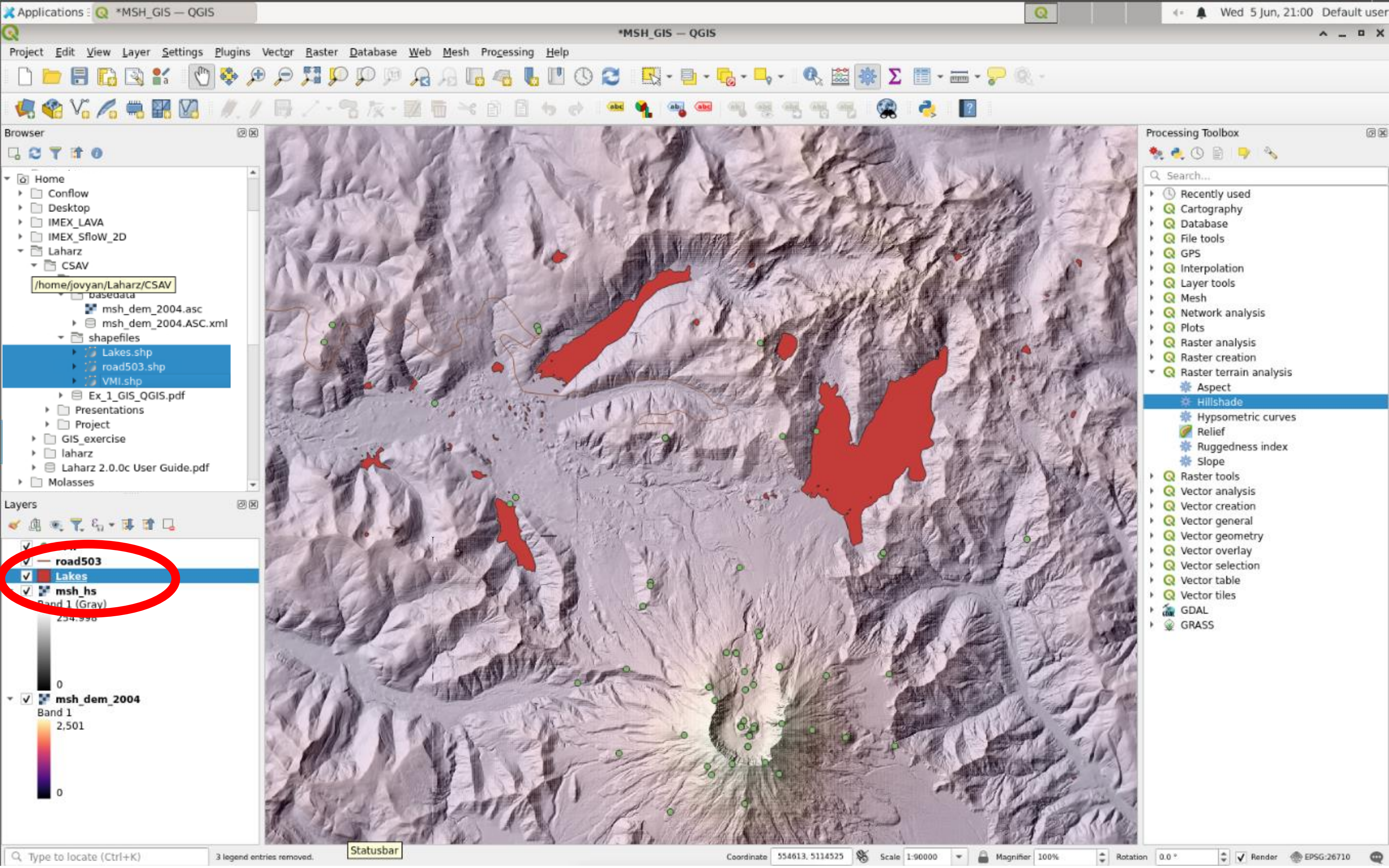
- Now let's make the hillshade slightly transparent
- Right click on "msh_hs" and select "Properties"
- Then select "Transparency" from the left menu
- Change the "Global opacity" to something less than 100% (I chose 85%)
- Click "OK"



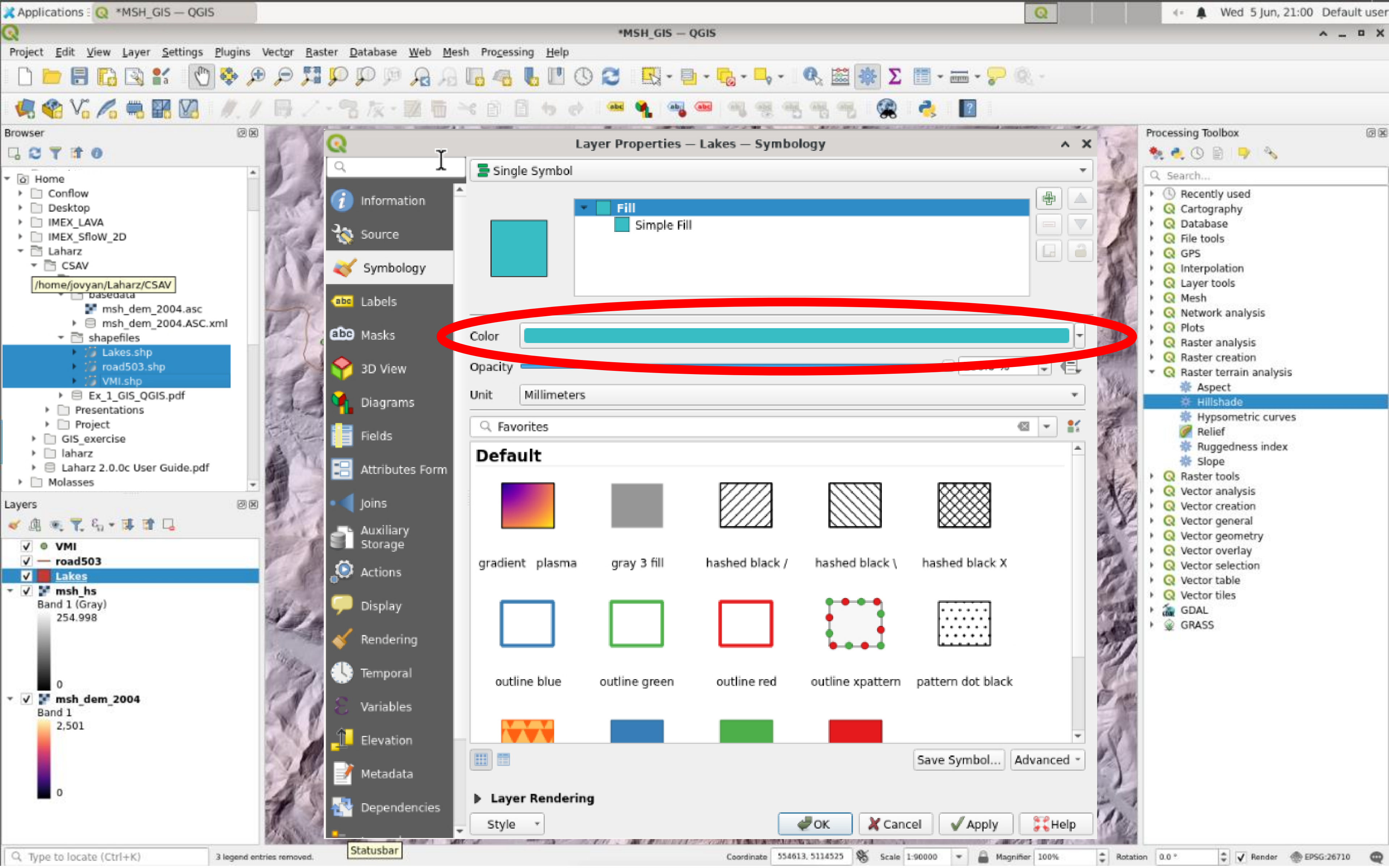
- Now’s a good time to save your project!
- Click the “Save” icon in the upper left



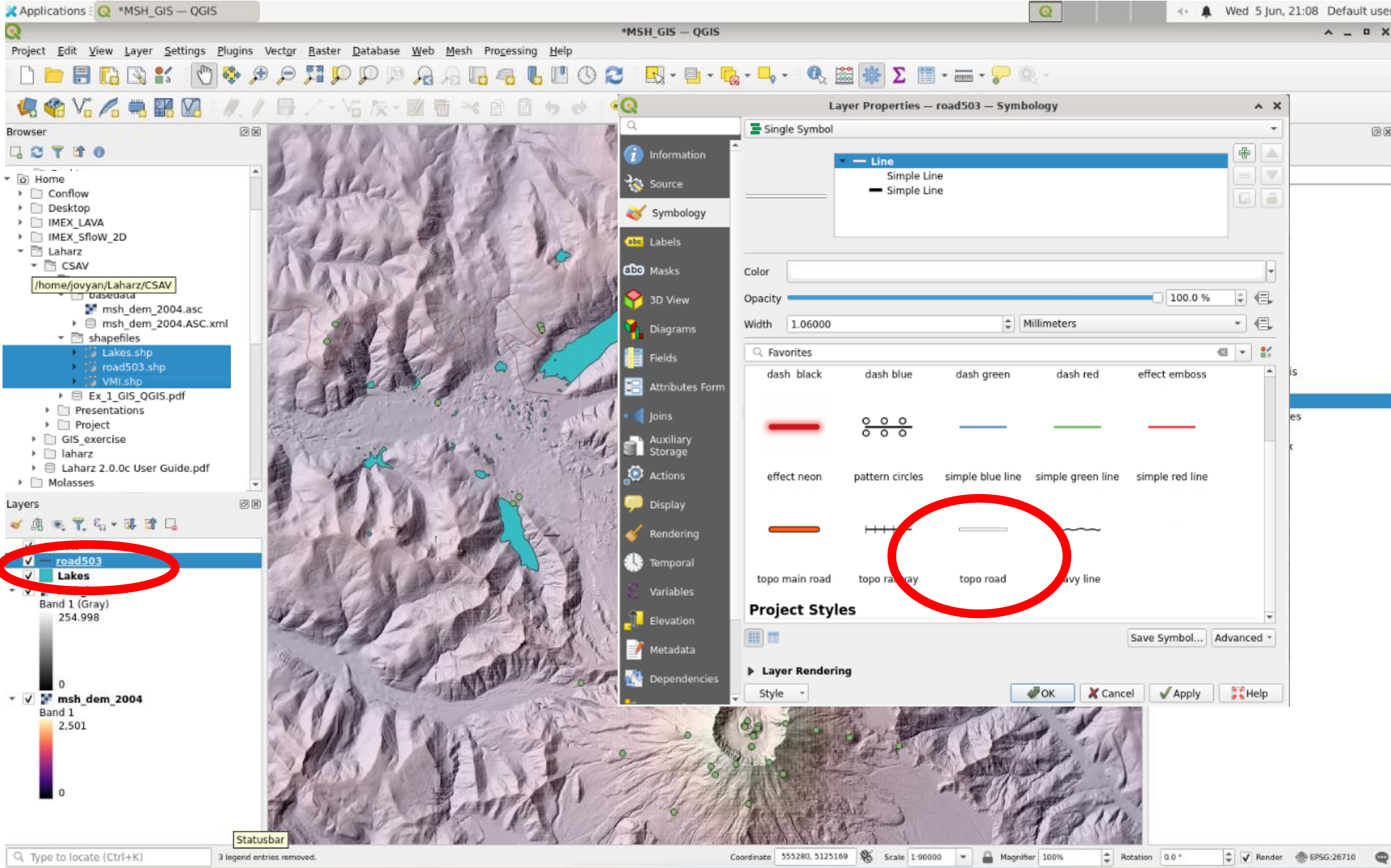
- Add data: Lakes, road503, and VMI shapefiles
- QGIS warns that these datasets have different coordinate systems
- You can leave the choice of transformations as the defaults



- The default symbology for these layers isn't very nice…
- Let's fix it!
- Double click on the "Lakes" layer to open the Layer Properties



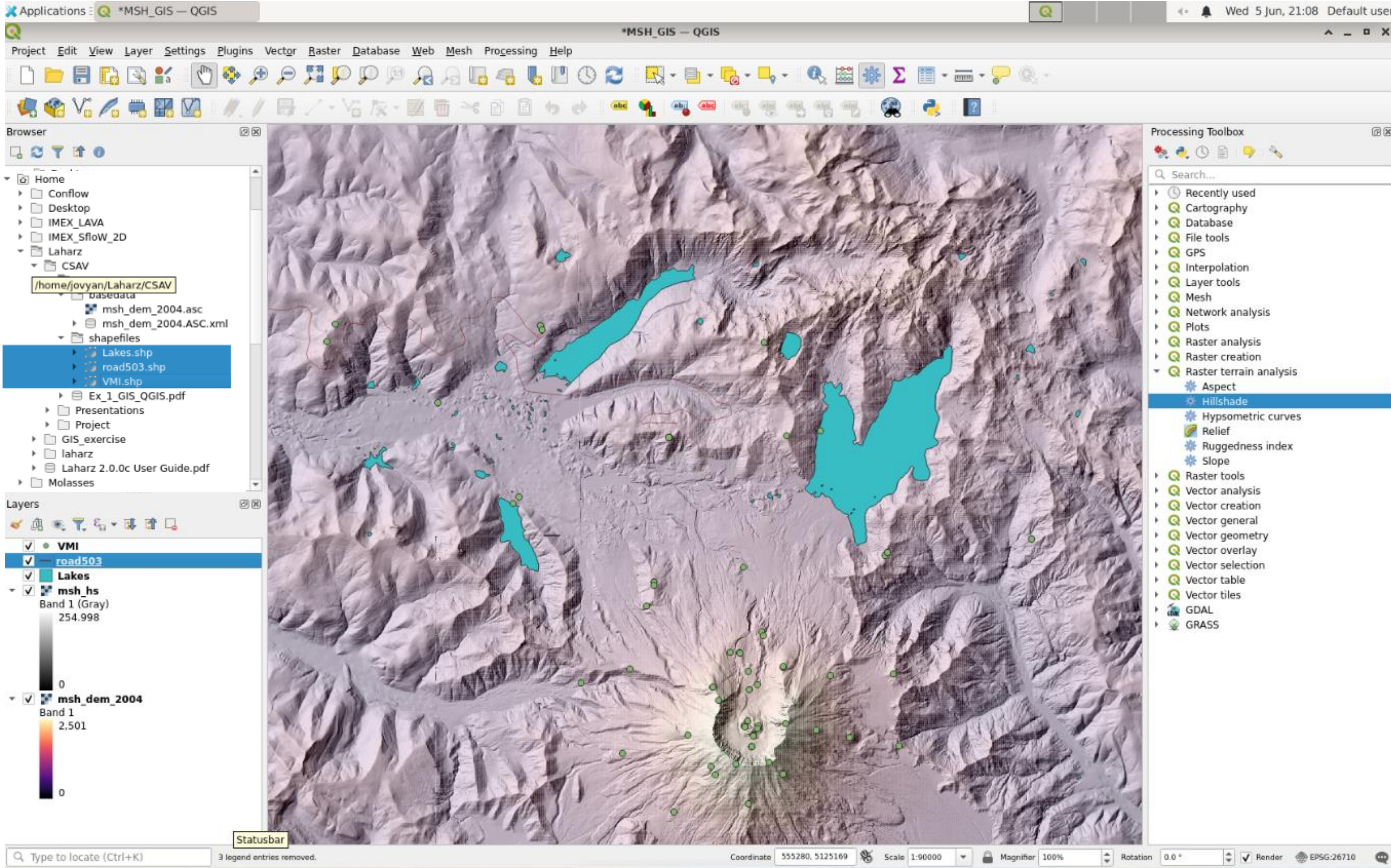
- Change the lake color to a lake color
- Use the drop-down menu labelled “color” and choose a color
- Click “Ok”



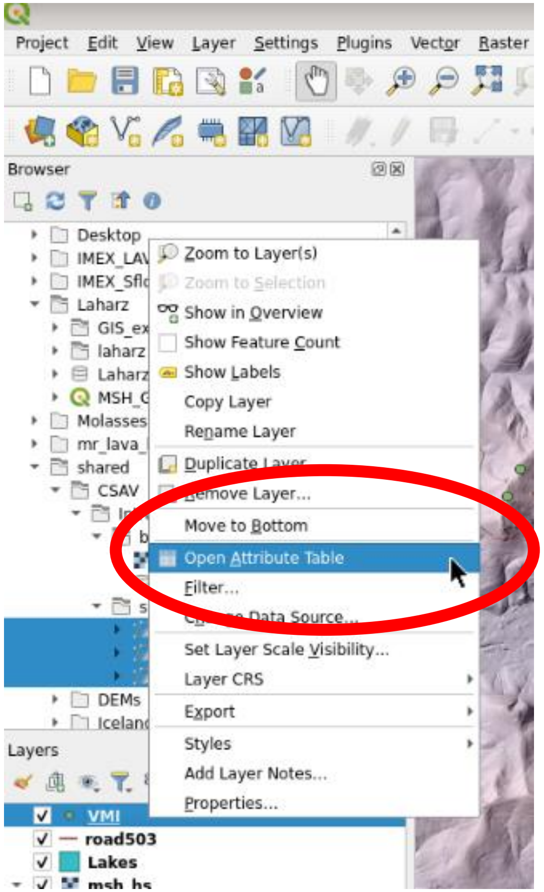
- The road is pretty hard to see. Let’s fix it.
- Open the layer properties for “road503” and change the symbology. I chose “topo road”
- Click “Ok”

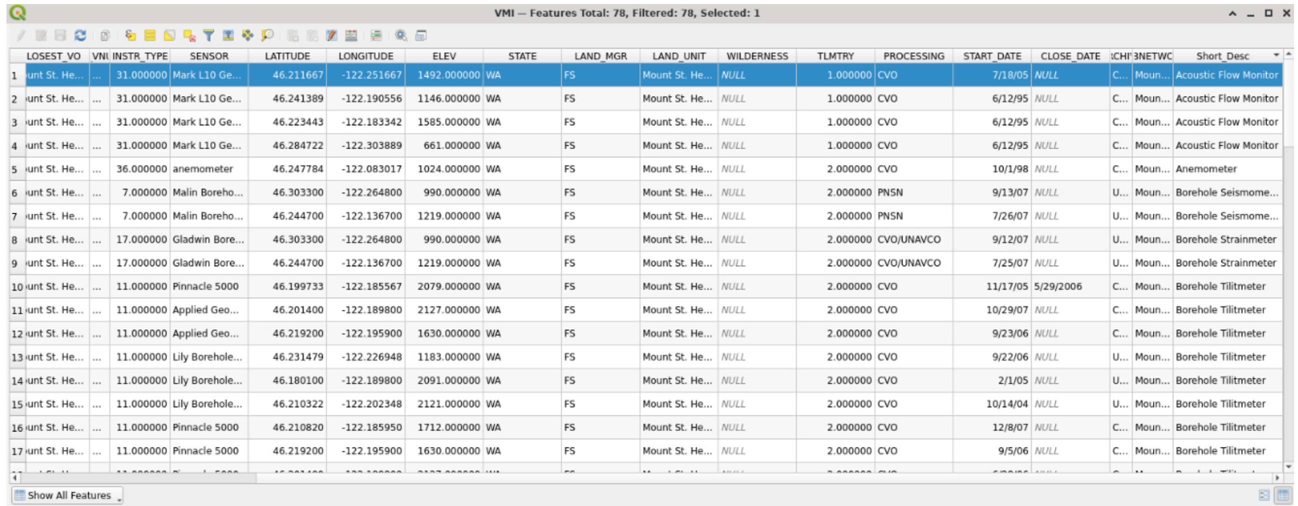
Exercise Goals

- Add Instrument data set and background information for context
- **View Instrument attribute table**
- Select instruments by drawing rectangle
- Select instruments by their attributes
- Assign colors to instrument type
- Add a legend to describe the meaning of colors



- The last layer is “VMI” or “volcano monitoring instruments”. It shows the network of instruments of different types.
- The default symbols are all the same, so we can’t tell what type of instruments these are. Let’s fix that!

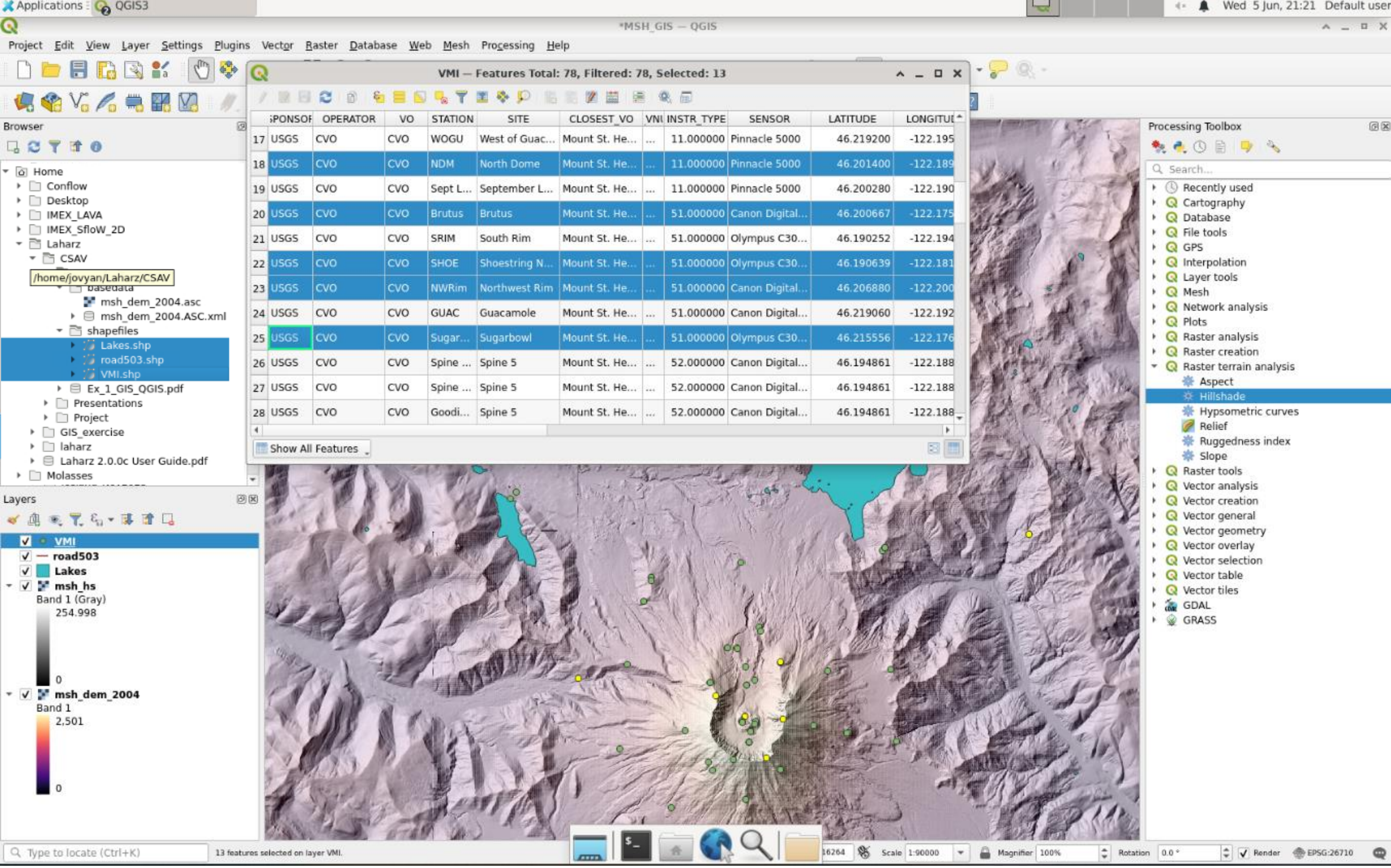



VMI — Features Total: 78, Filtered: 78, Selected: 1

	LOSEST_VO	VNI	INSTR_TYPE	SENSOR	LATITUDE	LONGITUDE	ELEV	STATE	LAND_MGR	LAND_UNIT	WILDERNESS	TLMTRY	PROCESSING	START_DATE	CLOSE_DATE	CHI	3NETWC	Short_Desc
1	unt St. He...	...	31.000000	Mark L10 Ge...	46.211667	-122.251667	1492.000000	WA	FS	Mount St. He...	NULL	1.000000	CVO	7/18/05	NULL	C...	Moun...	Acoustic Flow Monitor
2	unt St. He...	...	31.000000	Mark L10 Ge...	46.241389	-122.190556	1146.000000	WA	FS	Mount St. He...	NULL	1.000000	CVO	6/12/95	NULL	C...	Moun...	Acoustic Flow Monitor
3	unt St. He...	...	31.000000	Mark L10 Ge...	46.223443	-122.183342	1585.000000	WA	FS	Mount St. He...	NULL	1.000000	CVO	6/12/95	NULL	C...	Moun...	Acoustic Flow Monitor
4	unt St. He...	...	31.000000	Mark L10 Ge...	46.284722	-122.303889	661.000000	WA	FS	Mount St. He...	NULL	1.000000	CVO	6/12/95	NULL	C...	Moun...	Acoustic Flow Monitor
5	unt St. He...	...	36.000000	anemometer	46.247784	-122.083017	1024.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	10/1/98	NULL	C...	Moun...	Anemometer
6	unt St. He...	...	7.000000	Malin Boreho...	46.303300	-122.264800	990.000000	WA	FS	Mount St. He...	NULL	2.000000	PNSN	9/13/07	NULL	U...	Moun...	Borehole Seismome...
7	unt St. He...	...	7.000000	Malin Boreho...	46.244700	-122.136700	1219.000000	WA	FS	Mount St. He...	NULL	2.000000	PNSN	7/26/07	NULL	U...	Moun...	Borehole Seismome...
8	unt St. He...	...	17.000000	Gladwin Bore...	46.303300	-122.264800	990.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO/UNAVCO	9/12/07	NULL	U...	Moun...	Borehole Strainmeter
9	unt St. He...	...	17.000000	Gladwin Bore...	46.244700	-122.136700	1219.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO/UNAVCO	7/25/07	NULL	U...	Moun...	Borehole Strainmeter
10	unt St. He...	...	11.000000	Pinnacle 5000	46.199733	-122.185567	2079.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	11/17/05	5/29/2006	C...	Moun...	Borehole Tiltmeter
11	unt St. He...	...	11.000000	Applied Geo...	46.201400	-122.189800	2127.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	10/29/07	NULL	C...	Moun...	Borehole Tiltmeter
12	unt St. He...	...	11.000000	Applied Geo...	46.219200	-122.195900	1630.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	9/23/06	NULL	C...	Moun...	Borehole Tiltmeter
13	unt St. He...	...	11.000000	Lily Borehole...	46.231479	-122.226948	1183.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	9/22/06	NULL	U...	Moun...	Borehole Tiltmeter
14	unt St. He...	...	11.000000	Lily Borehole...	46.180100	-122.189800	2091.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	2/1/05	NULL	U...	Moun...	Borehole Tiltmeter
15	unt St. He...	...	11.000000	Lily Borehole...	46.210322	-122.202348	2121.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	10/14/04	NULL	U...	Moun...	Borehole Tiltmeter
16	unt St. He...	...	11.000000	Pinnacle 5000	46.210820	-122.185950	1712.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	12/8/07	NULL	C...	Moun...	Borehole Tiltmeter
17	unt St. He...	...	11.000000	Pinnacle 5000	46.219200	-122.195900	1630.000000	WA	FS	Mount St. He...	NULL	2.000000	CVO	9/5/06	NULL	C...	Moun...	Borehole Tiltmeter

Show All Features

- Right click on “VMI” and choose “Open Attribute Table”
- We can see all kinds of information about the sensors, including the sensor type (“Short_Desc” column all the way to the right)

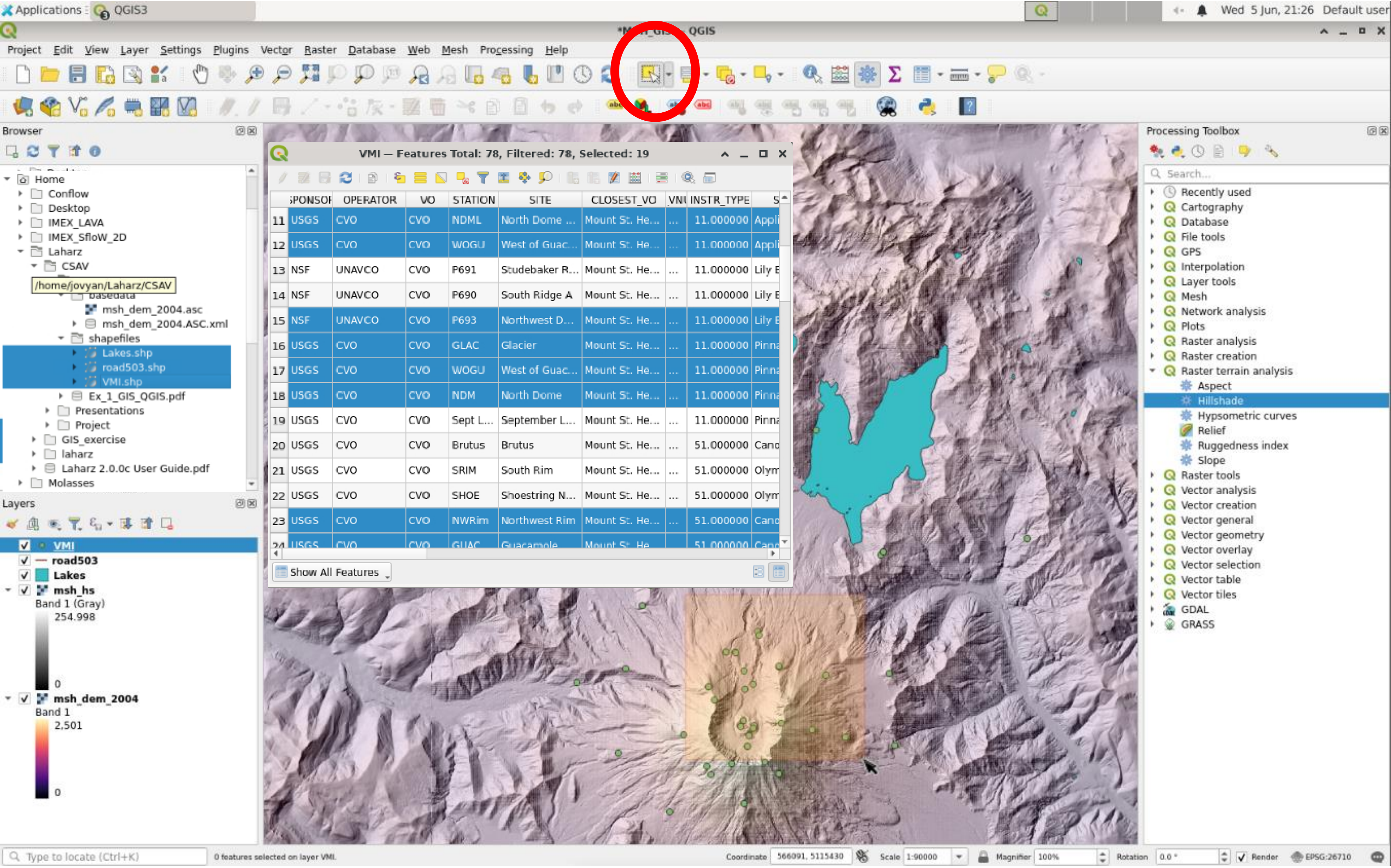


	SPONSOR	OPERATOR	VO	STATION	SITE	CLOSEST_VO	VNI	INSTR_TYPE	SENSOR	LATITUDE	LONGITUDE
17	USGS	CVO	CVO	WOGU	West of Guac...	Mount St. He...	...	11.000000	Pinnacle 5000	46.219200	-122.195
18	USGS	CVO	CVO	NDM	North Dome	Mount St. He...	...	11.000000	Pinnacle 5000	46.201400	-122.189
19	USGS	CVO	CVO	Sept L...	September L...	Mount St. He...	...	11.000000	Pinnacle 5000	46.200280	-122.190
20	USGS	CVO	CVO	Brutus	Brutus	Mount St. He...	...	51.000000	Canon Digital...	46.200667	-122.175
21	USGS	CVO	CVO	SRIM	South Rim	Mount St. He...	...	51.000000	Olympus C30...	46.190252	-122.194
22	USGS	CVO	CVO	SHOE	Shoestring N...	Mount St. He...	...	51.000000	Olympus C30...	46.190639	-122.181
23	USGS	CVO	CVO	NWRim	Northwest Rim	Mount St. He...	...	51.000000	Canon Digital...	46.206880	-122.200
24	USGS	CVO	CVO	GUAC	Guacamole	Mount St. He...	...	51.000000	Canon Digital...	46.219060	-122.192
25	USGS	CVO	CVO	Sugar...	Sugarbowl	Mount St. He...	...	51.000000	Olympus C30...	46.215556	-122.176
26	USGS	CVO	CVO	Spine ...	Spine 5	Mount St. He...	...	52.000000	Canon Digital...	46.194861	-122.188
27	USGS	CVO	CVO	Spine ...	Spine 5	Mount St. He...	...	52.000000	Canon Digital...	46.194861	-122.188
28	USGS	CVO	CVO	Goodi...	Spine 5	Mount St. He...	...	52.000000	Canon Digital...	46.194861	-122.188

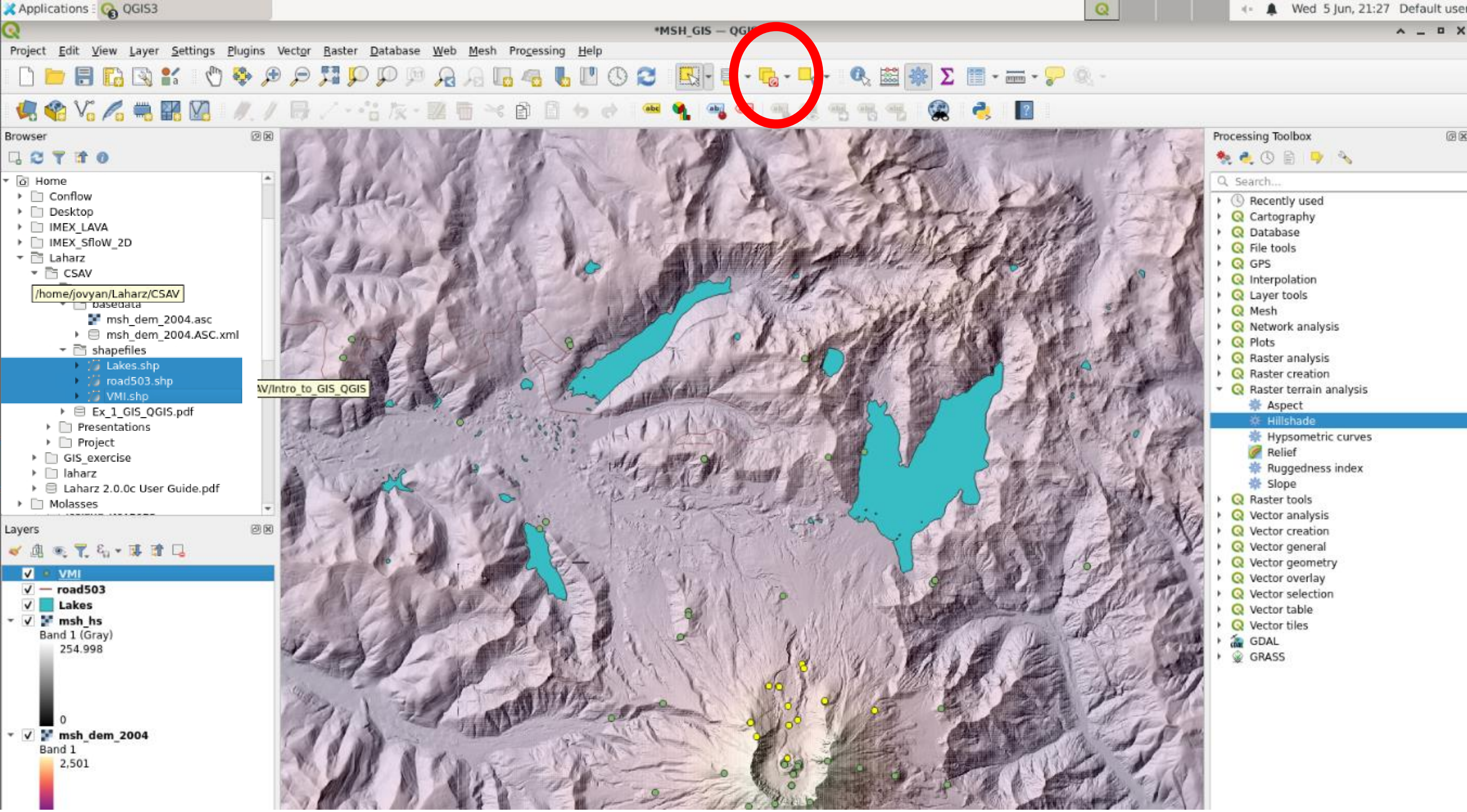
- When you select rows in the attribute table (Ctrl+Click the row number), you will see those instruments turn yellow on the map
- Close the attribute table

Exercise Goals

- Add Instrument data set and background information for context
- View Instrument attribute table
- **Select instruments by a drawing rectangle**
- Select instruments by their attributes
- Assign colors to instrument type
- Add a legend to describe the meaning of colors



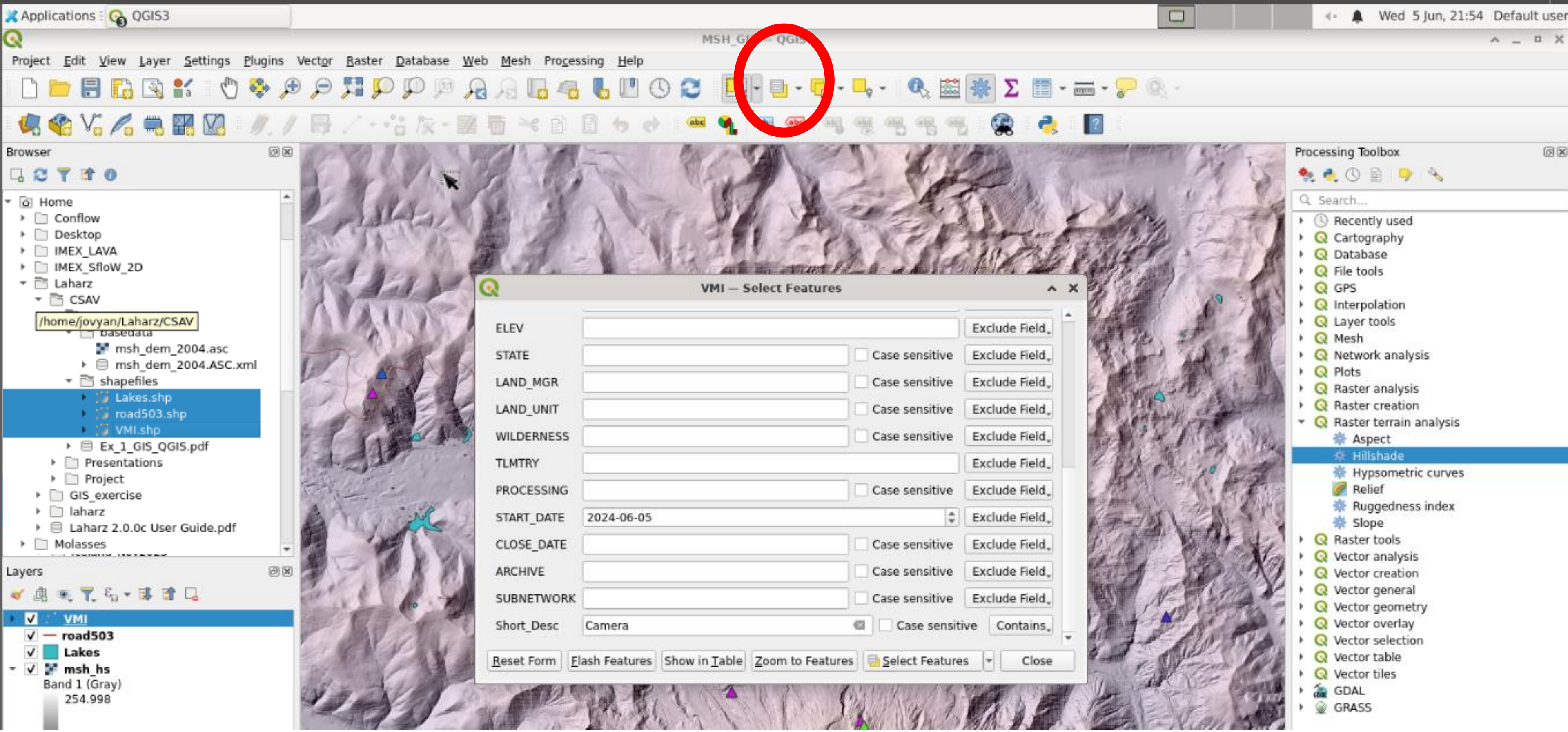
- Choose the “Select” tool from the menu
- Draw a rectangle around some of the VMI stations
- The selected stations will highlight in yellow
- If you open the attribute table you will see the selected stations highlighted



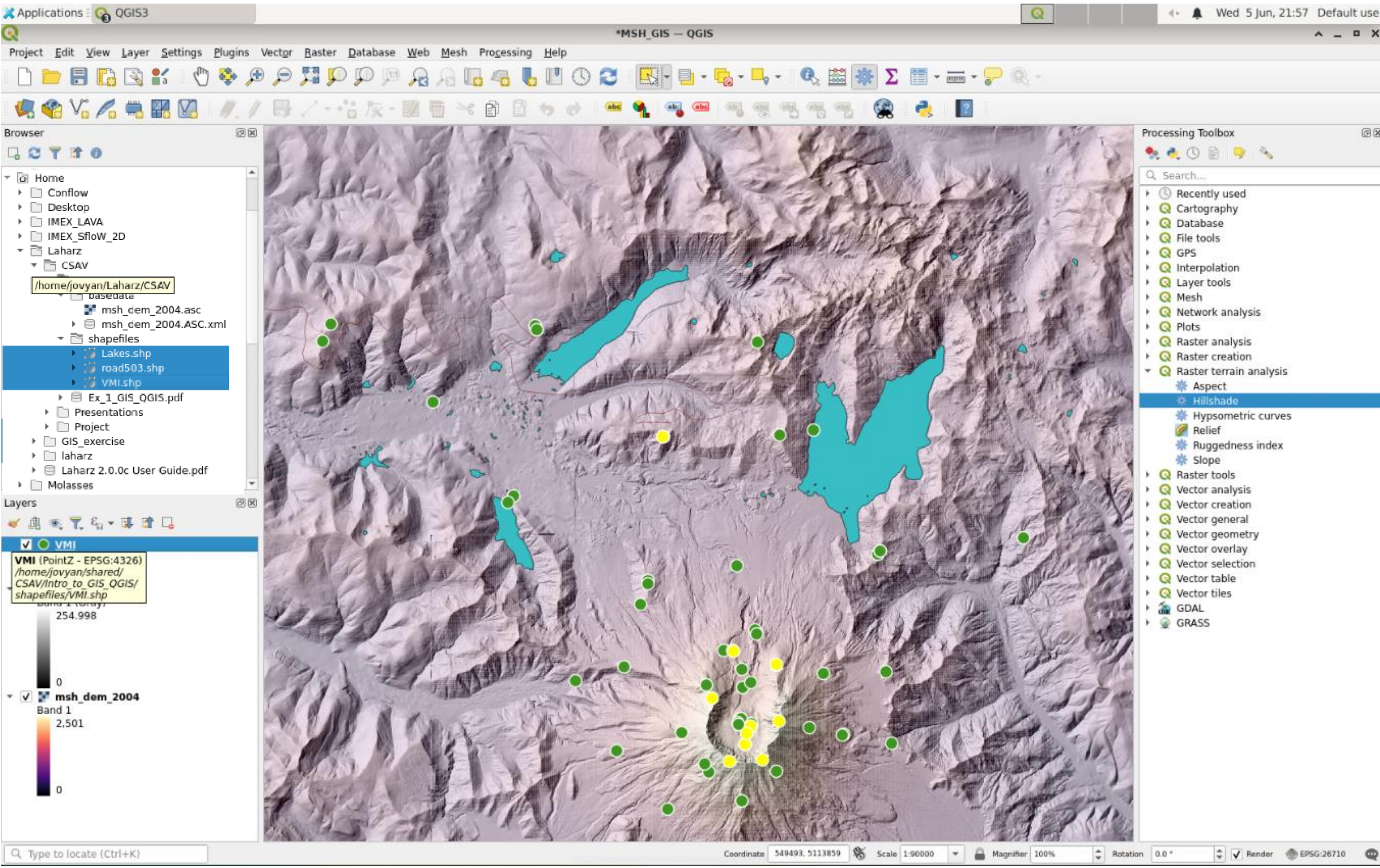
- To clear the selection, click the “Deselect features from all layers” button

Exercise Goals

- Add Instrument data set and background information for context
- View Instrument attribute table
- Select instruments by a drawing rectangle
- **Select instruments by their attributes**
- Assign colors to instrument type
- Add a legend to describe the meaning of colors



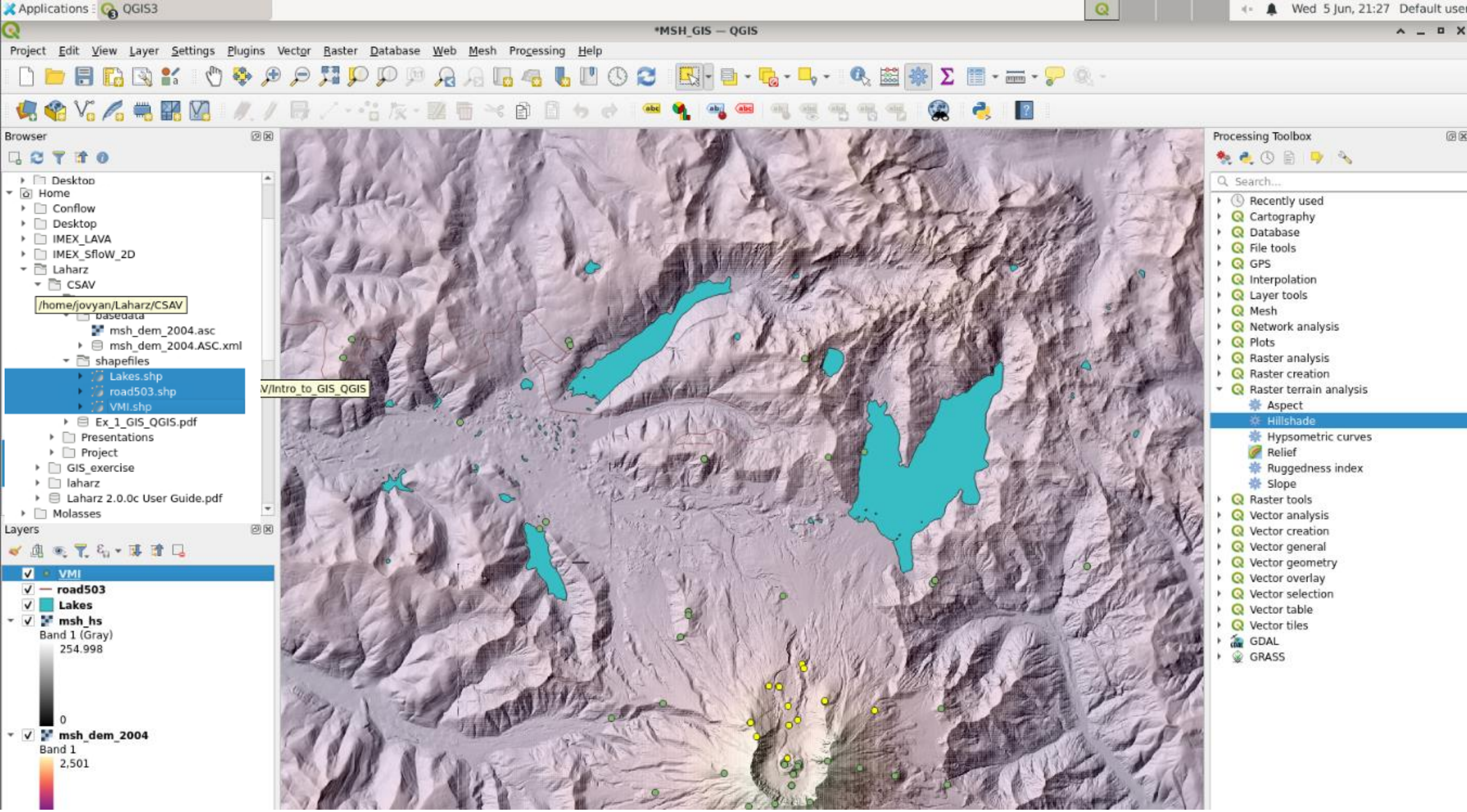
- We can query the attribute table to select certain features based on data from the table
- Select the “Select features by value” tool
- Scroll down until you see “Short_Desc”
- Change the button that says “Exclude Field” to “Contains”
- Type “Camera” in the text box
- Click “Select Features”



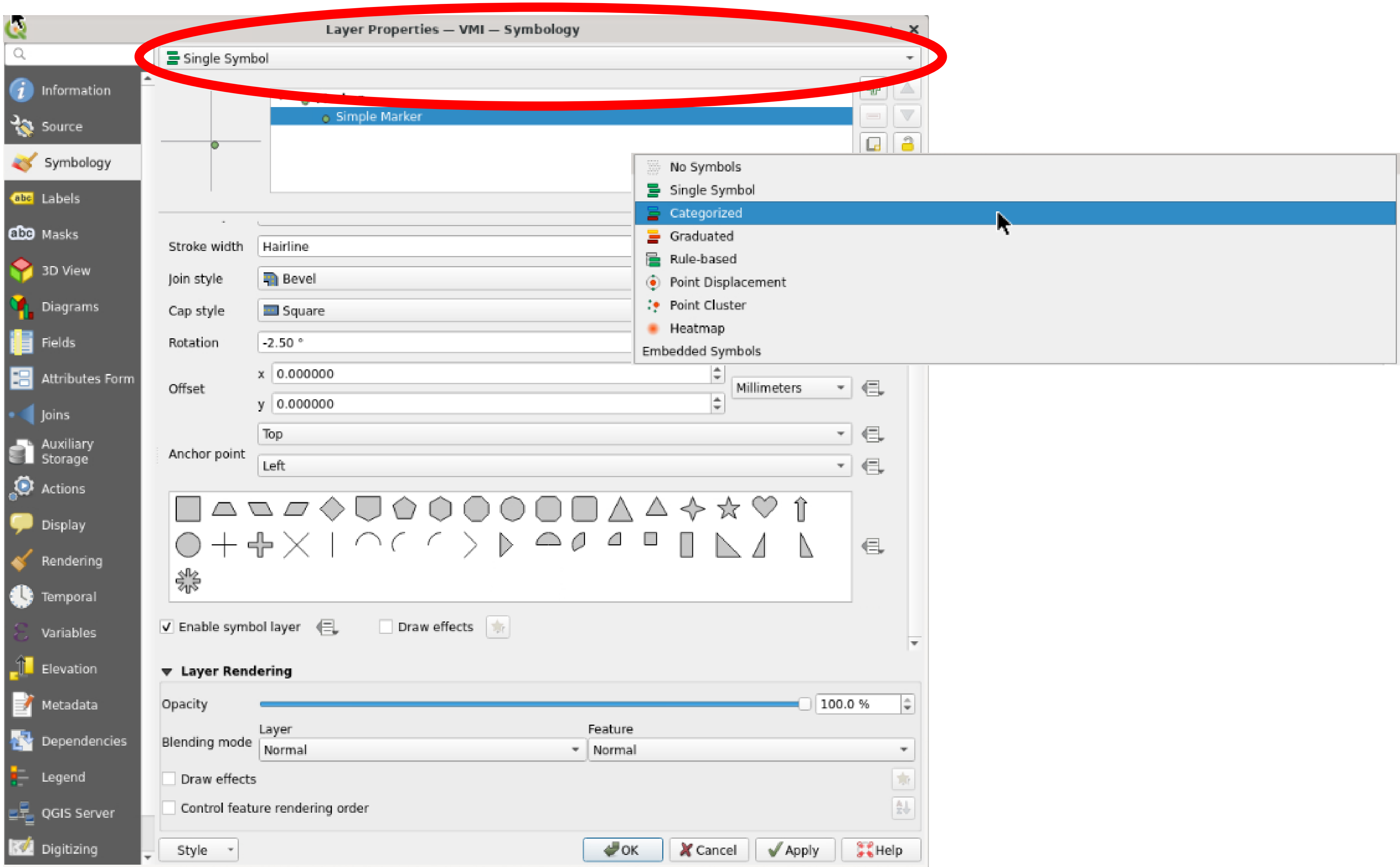
- The camera stations will be highlighted in yellow

Exercise Goals

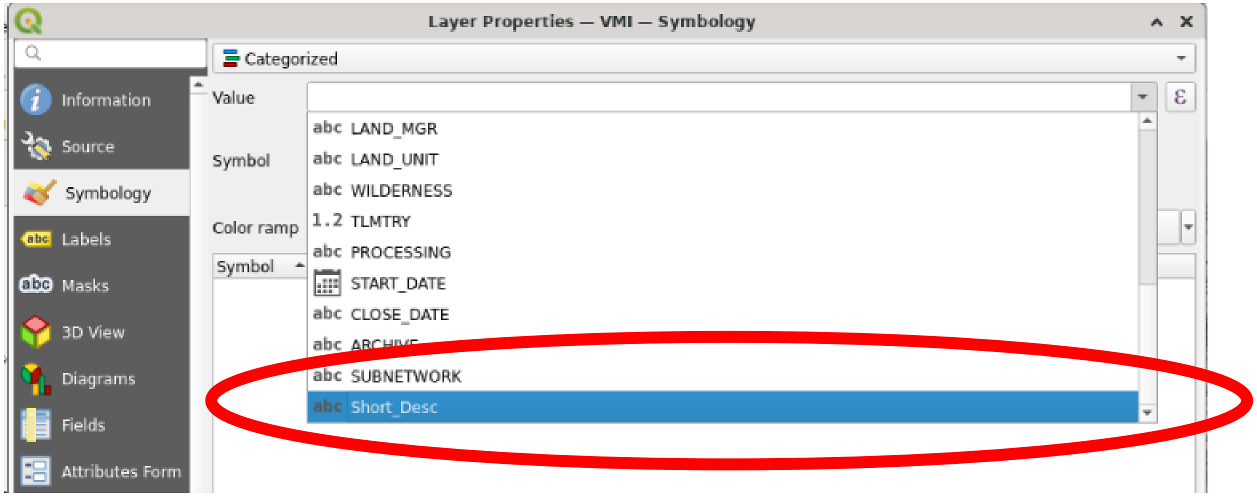
- Add Instrument data set and background information for context
- View Instrument attribute table
- Select instruments by a drawing rectangle
- Select instruments by their attributes
- **Assign colors to instrument type**
- Add a legend to describe the meaning of colors



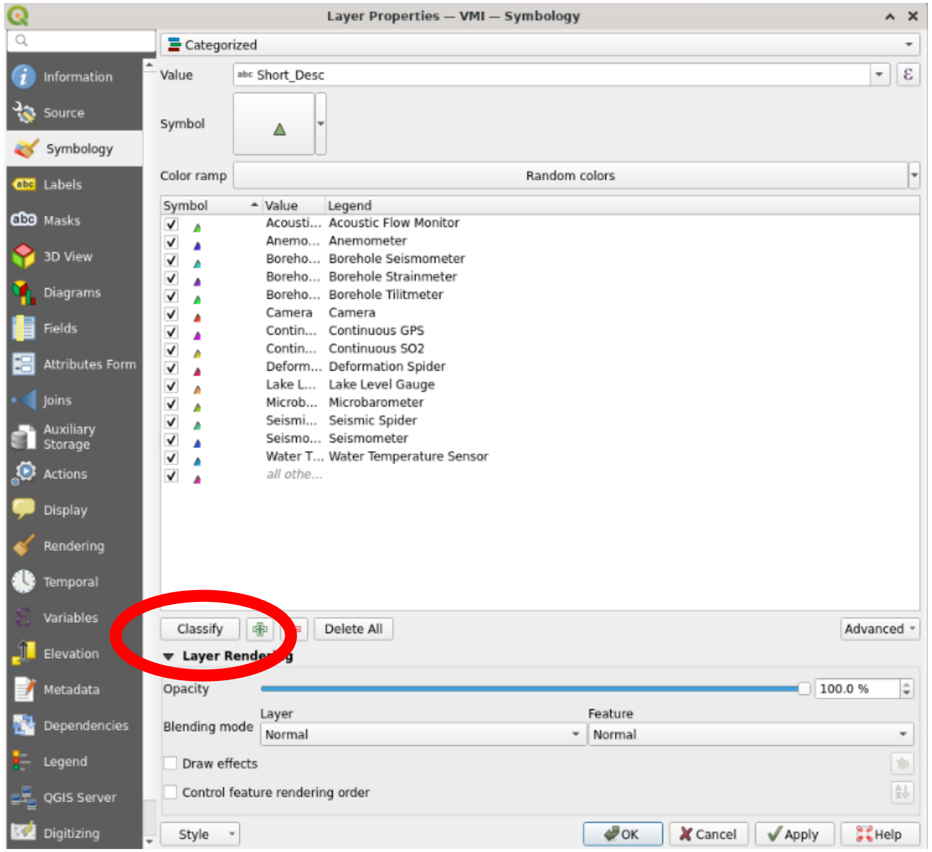
- We can use the attribute table to assign different symbols to the VMI layer so we can more easily tell what kind of stations these are
- Double click the VMI layer and open the Layer Properties

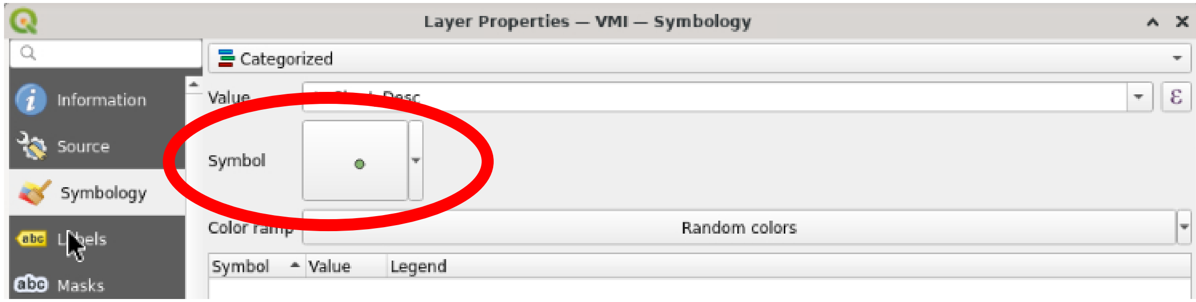


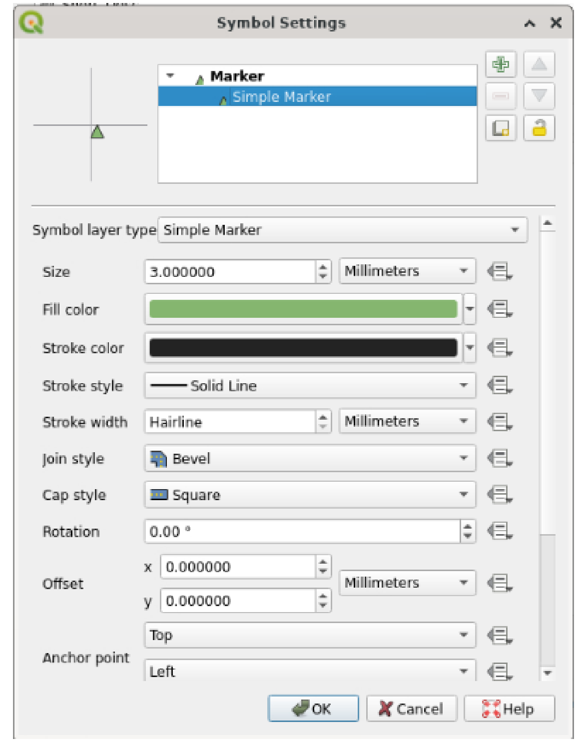
- Change from “Single symbol” to “Categorized”



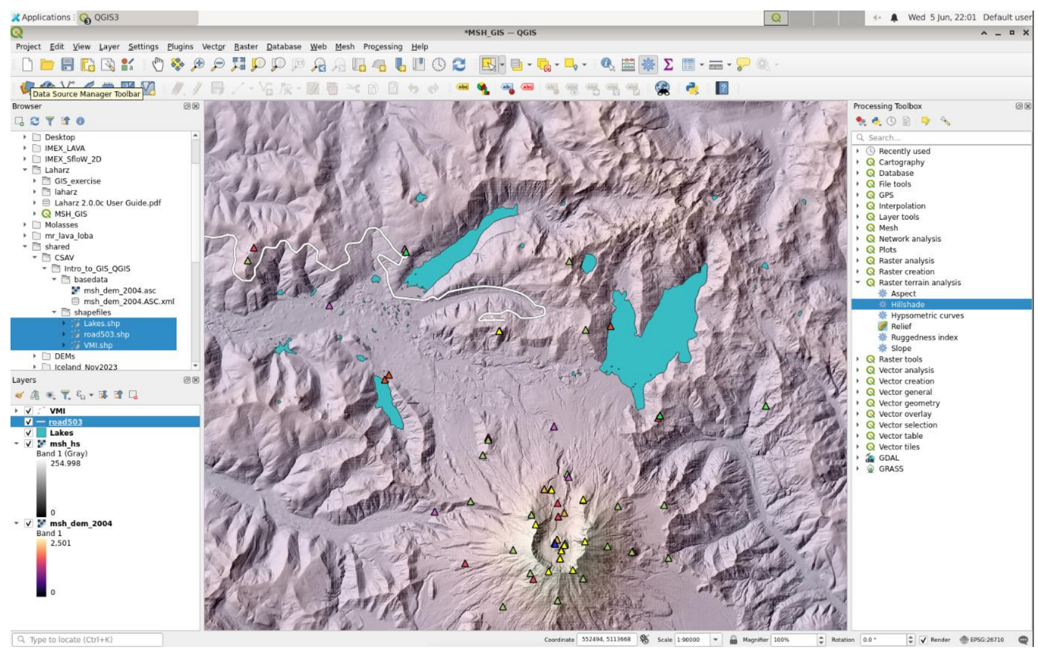
- Change the "Value" to the column you want to use for the display. We want to know what kind of stations these are, so choose "Short_desc".
- Then click "classify" and all the different station types should appear





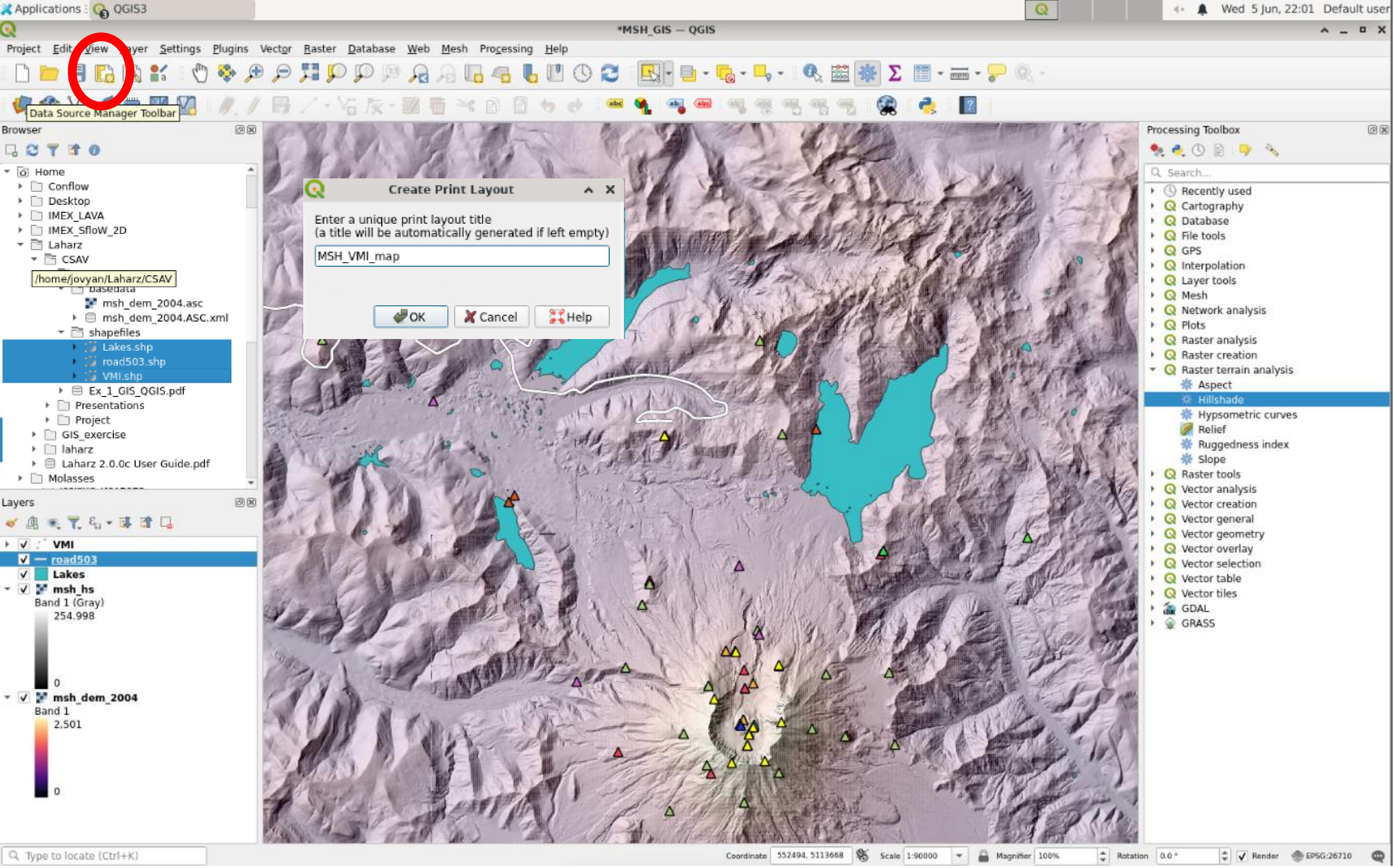


- Click the “Symbol” button to change the size and shape of the symbols. I chose 3 mm triangles
- Because station type is a categorical variable, choose “Random colors” for color.
- You can also change the colors for each symbol if you wanted
- Click “OK”
- Remember to save your project!

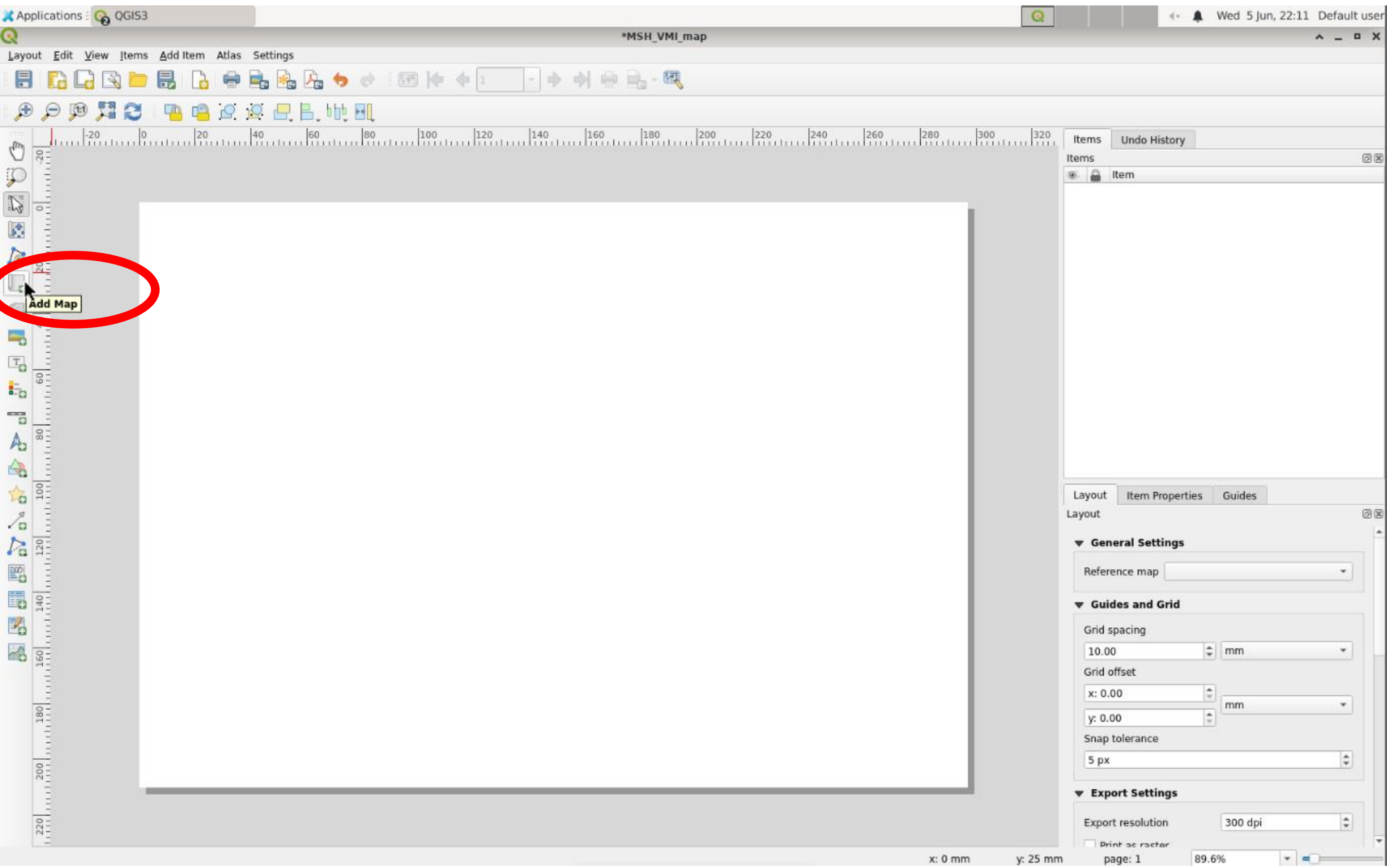


Exercise Goals

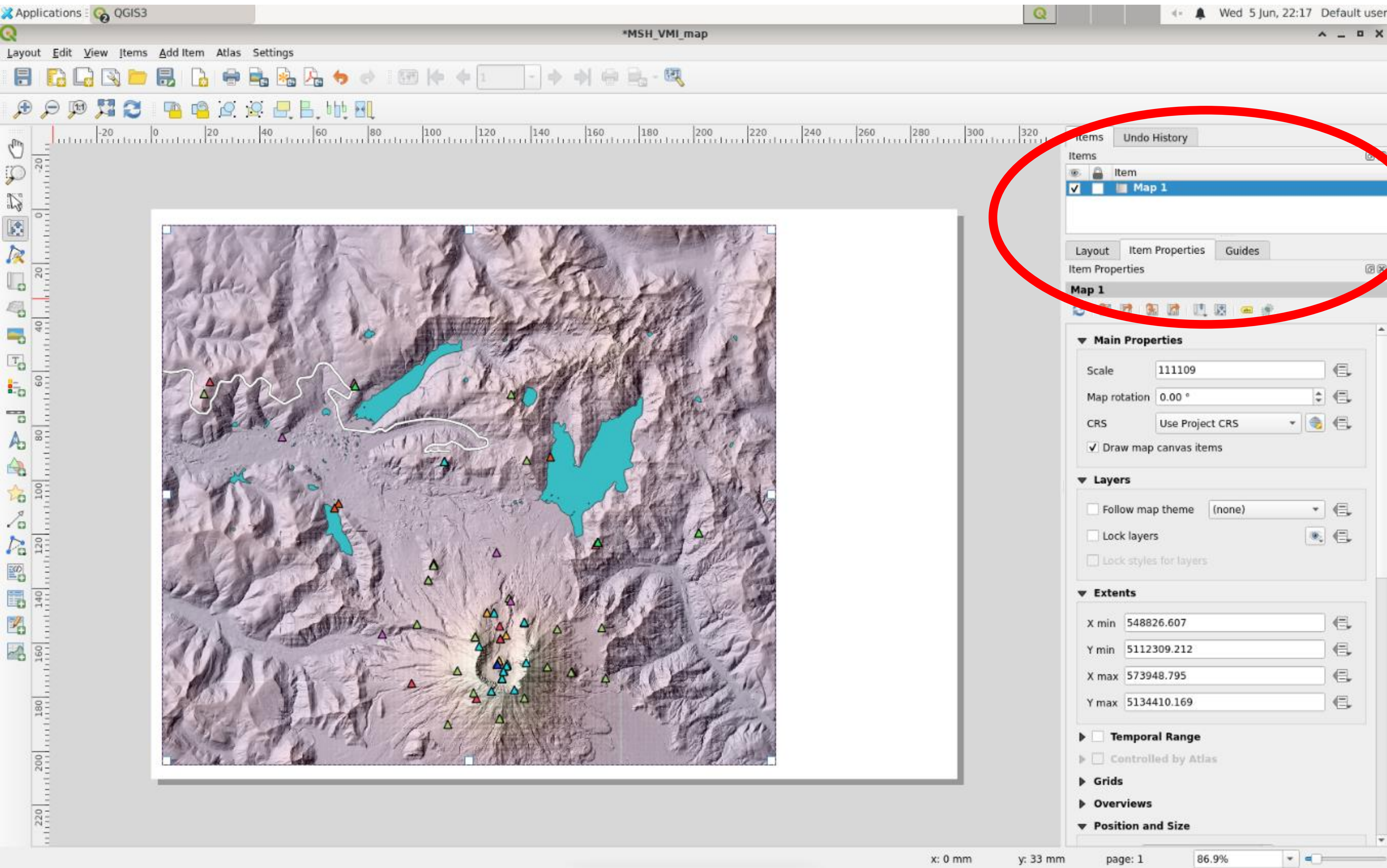
- Add Instrument data set and background information for context
- View Instrument attribute table
- Select instruments by a drawing rectangle
- Select instruments by their attributes
- Assign colors to instrument type
- **Add a legend to describe the meaning of colors**



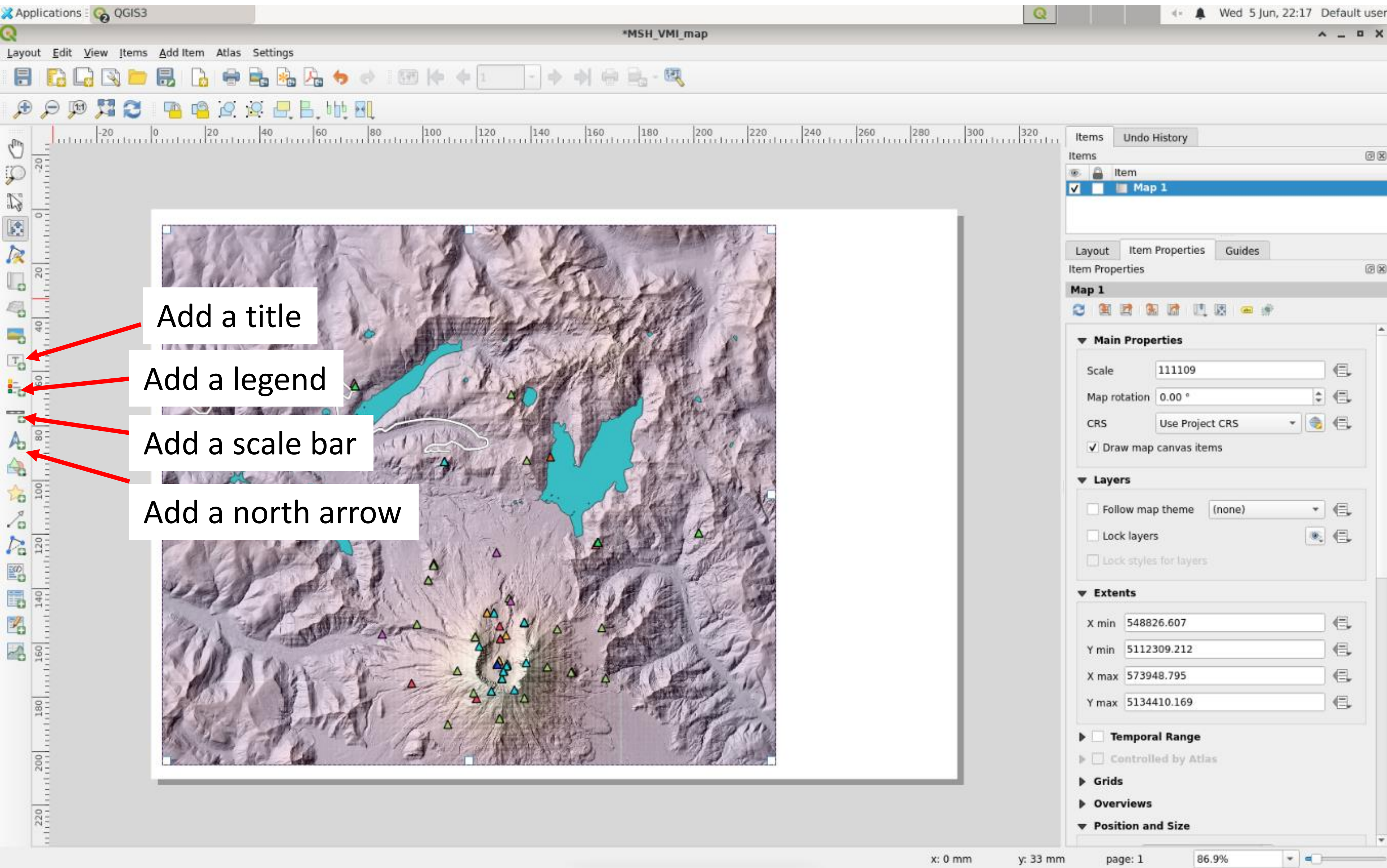
- Let’s make a map!
- Click the “New Print Layout” button
- Enter a name for the new map. I chose “MSH_VMI_map”
- Click “OK”



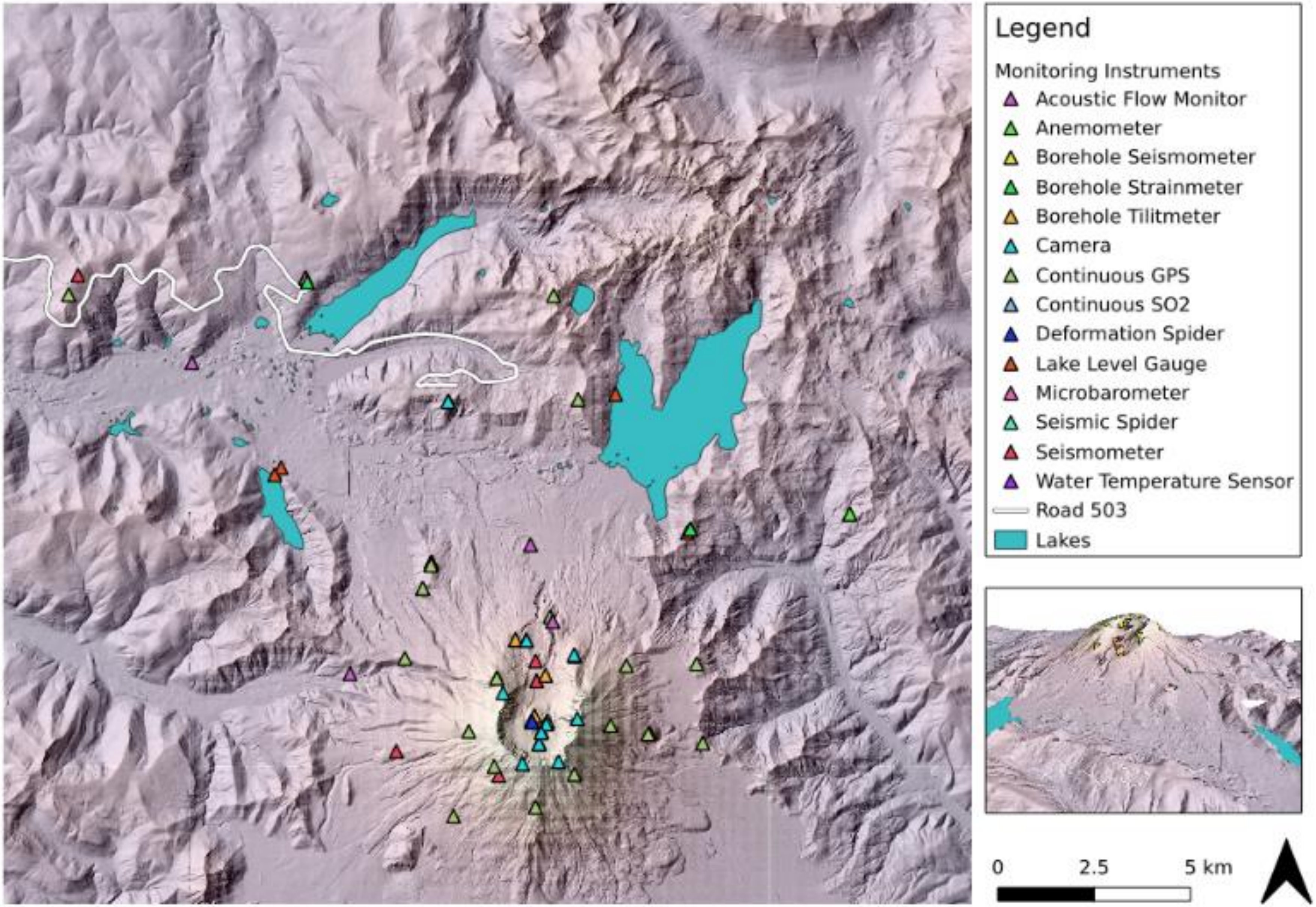
- First you need to add your map to the layout
- Click the “Add map” button
- Then draw a rectangle on the blank page to add the map



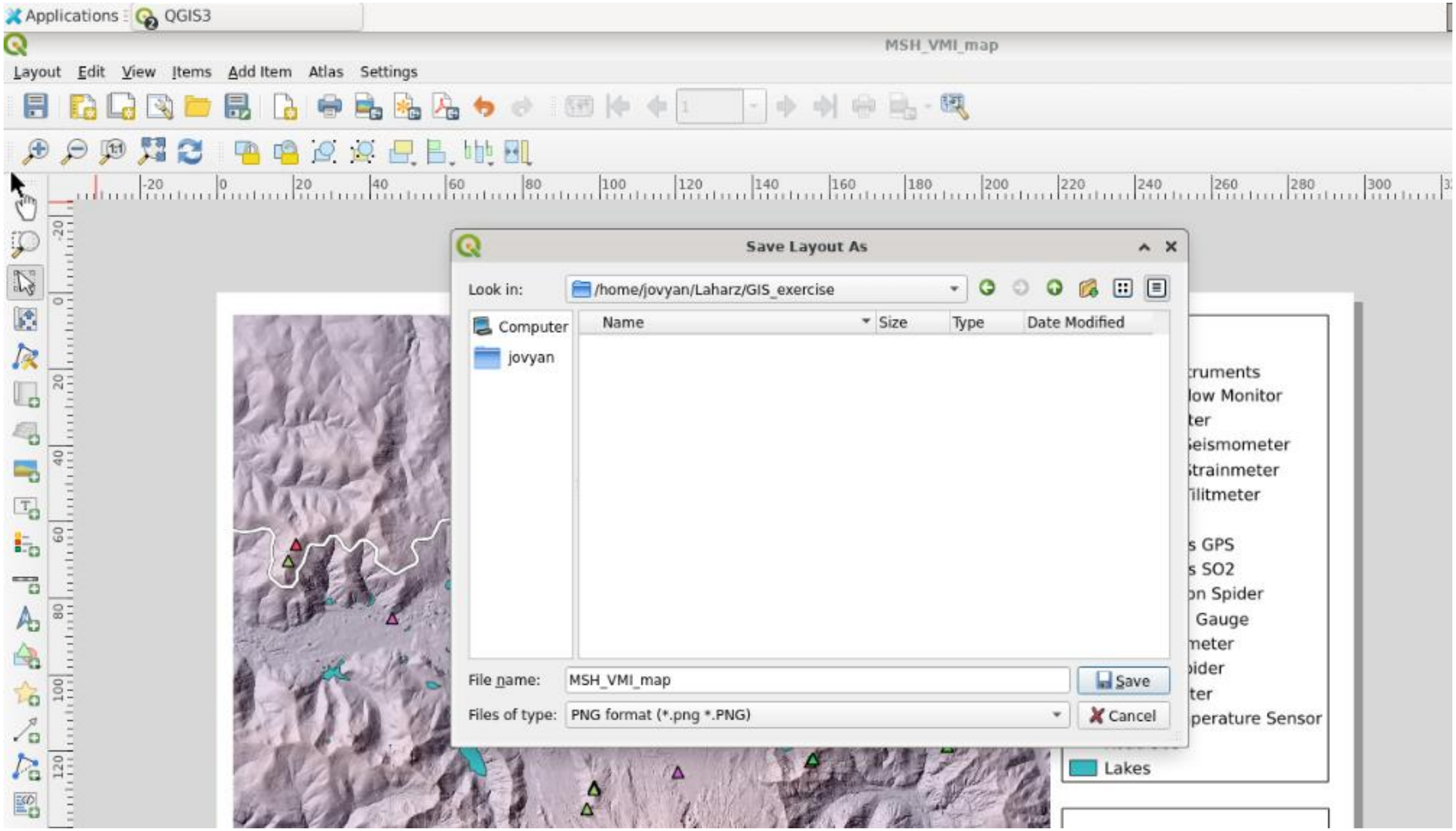
- You can adjust the scale and extent and other properties by selecting the map layer and "Item properties"



- You can add a legend for the stations and other layers
- Add a scale bar and/pr coordinates
- Add a north arrow
- Play around with the options to make your map



- Save your final work!
- You can also export the map as an image



- Save your final work!
- You can also export the map as an image